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AIR FORCE TEST CENTER

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This publication implements Air Force Instruction (AFI) 91-202 Air Force Materiel Command (AFMC) Supplement, *The US Air Force Mishap prevention Program.* This publication provides further policy and guidance to **Chapter 13**. It directs the application of system safety principles to the planning and conduct of all Air Force Test Center (AFTC) and other designated AFMC test projects (reference **paragraph 1.6**) regardless of the agency conducting the tests. It also provides guidance for the application of system safety principles to AFTC training programs, logistics testing, and publications. Organizations within AFTC will supplement this instruction to provide a detailed local test safety review process. All direct Supplements must be routed to the Office of Primary Responsibility (OPR) of this publication for review and approval prior to certification and approval by the 412 and 96 Test Wings (TWs) or Arnold Engineering Development Complex (AEDC) Commander. **Attachment 1** lists abbreviations and acronyms used in this instruction. Refer recommended changes and questions about this publication to the

OPR using the AF Form 847, *Recommendation for Change of Publication*; route AF Forms 847 from the field through the appropriate functional chain of command. The authority to waive wing/unit level requirements in this publication is Tier 3. See AFI 33-360, *Publications and Forms Management*, Table 1.1 for a description of the authorities associated with the Tier numbers. Submit requests for waivers through the chain of command to the appropriate Tier waiver approval authority, or alternately, to the Publication OPR for non-tiered compliance items. Ensure that all records created as a result of processes prescribed in this publication are maintained in accordance with (IAW) Air Force Manual (AFMAN) 33-363, *Management of Records*, and disposed of IAW Air Force Records Disposition Schedule (RDS) located in the Air Force Records Information Management System (AFRIMS). The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Air Force.

(96TWSUP) This supplement implements Air Force Test Center Instruction 91-202. This supplement applies to all activities conducted through the 96th Test Wing (96 TW) including, but not limited to, test and training operations. The term "test" as it relates to the safety review process includes both test and training operations conducted through the 96 TW. This supplement provides clarification to responsibilities and procedures for test safety reviews for the 96 TW. Tests include ground, waterborne and airborne testing and training activities involving 96 TW personnel, aircraft, test ranges, equipment or airspace. It applies to system program managers, program engineers (or equivalent), test engineers, test and range safety engineers and aircrews that are responsible for incorporating safety planning and review into the conduct of test and training programs. Refer recommended changes and questions about this publication to the Office of Primary Responsibility (OPR) using the AF Form 847, *Recommendation for Change of Publication*; route AF Forms 847 from the field through the appropriate functional's chain of command. Ensure that all records created as a result of processes prescribed in this publication are maintained IAW Air Force Instruction (AFI) 33-322, *Records Management and Information Governance Program*, and disposed of IAW Air Force Records Information System (AFRIMS) Records Disposition Schedule (RDS).

SUMMARY OF CHANGES

The major changes to this instruction include: providing the option for the SRB Chair's signature to represent the signature of all SRB members; removal of Wing/Complex Test Safety Office approval of T-2 modifications and requiring test teams to review those modifications; permission to use personnel from within the test unit as independent safety reviewers; redefinition of the Remote and Improbable mishap probabilities; removal of the lines of subjectivity in the risk assessment matrix; change to the approval authority for Low Risk tests from the Group Commander to the Squadron Commander; rewrite of the unexpected test event section; permission to use memorandums for approving minor safety plan changes and major test plan changes with no change to safety planning; and correction to the definition of mishap accountability.

Chapter 1—INTRODUCTION 6

1.1. General. 6

1.2. Test Safety Review Process. 6

Figure 1.1. AFTCI 91-202 Process Flow. 7

1.3. Safety Review Process Goals. 8

1.4. Risk Management 8

1.5. Safety Mindset. 9

1.6. Scope. 9

1.7. Waivers to This Instruction. 9

1.8. Authority. 9

Chapter 2—SAFETY RESPONSIBILITIES 11

2.1. Test Safety Approval and Coordination Responsibilities 11

2.2. Test Unit Safety Planning Responsibilities. 13

2.3. Independent Safety Reviewer Responsibilities 16

Chapter 3—TEST SAFETY PLANNING PHASE 19

3.1. Test and Safety Planning. 19

3.2. Safety Considerations during Test Planning 19

3.3. Safety Planning Objectives. 19

3.4. Test Package Documentation 21

Chapter 4—RISK ASSESSMENT 28

4.1. General. 28

4.2. Determine Mishap Severity. 28

Table 4.1. Mishap Severity Definitions. 29

4.3. Determine Mishap Probability. 29

Table 4.2. Mishap Probability Definitions. 30

4.4. Risk Assessment Matrix. 30

Figure 4.1. Risk Assessment Matrix. 31

4.5. NEGLIGIBLE Risk. 31

4.6. Determine Overall Risk Assessment. 32

Table 4.3. Overall Risk Level Assessments. 33
4.7. Elevated Risk Activities. 33
Chapter 5—TEST SAFETY REVIEW PHASE **34**
5.1. Safety Review Preparation. 34
5.2. Safety Review. 34
Table 5.1. (Added-96TWSUP) SRB Timeline. 37
Chapter 6—TEST SAFETY APPROVAL PHASE **45**
6.1. Approval Authorities and Notification Levels. 45
Table 6.1. Approval Process Coordination Path. 45
6.2. LOW Risk Activities. 46
6.3. Elevated Risk Activities. 46
6.4. Test Approval Brief. 46
6.5. Acceptance of Safety Planning across AFTC 47
Chapter 7—TEST EXECUTION PHASE **48**
7.1. General. 48
7.2. Test Card/Test Period Directive Preparation and Approval 48
7.3. Test/Mission Execution Briefing. 49
7.4. Unexpected Test Event 49
7.5. Hazard Occurrence but Not an Unexpected Test Event. 49
Chapter 8—CHANGES AND TIME LIMITS **50**
8.1. Changes. 50
8.2. Major Changes. 50
8.3. Minor and Administrative Changes. 51
8.4. Time Limit. 51
8.5. Closure Amendments/Lessons Learned. 52
Attachment 1—GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION **53**
Attachment 2—(Added-96TWSUP) SAMPLE SAFETY APPENDIX FORMAT **59**
Attachment 3—(Added-96TWSUP) INSTRUCTIONS FOR COMPLETING AFTC FORM 5000, TEST HAZARD ANALYSIS FORM **61**

Attachment 4—(Added-96TWSUP) SAMPLE SAFETY REVIEW BOARD SUMMARY 63

Attachment 5—(Added-96TWSUP) SAMPLE LESSONS LEARNED WORKSHEET 65

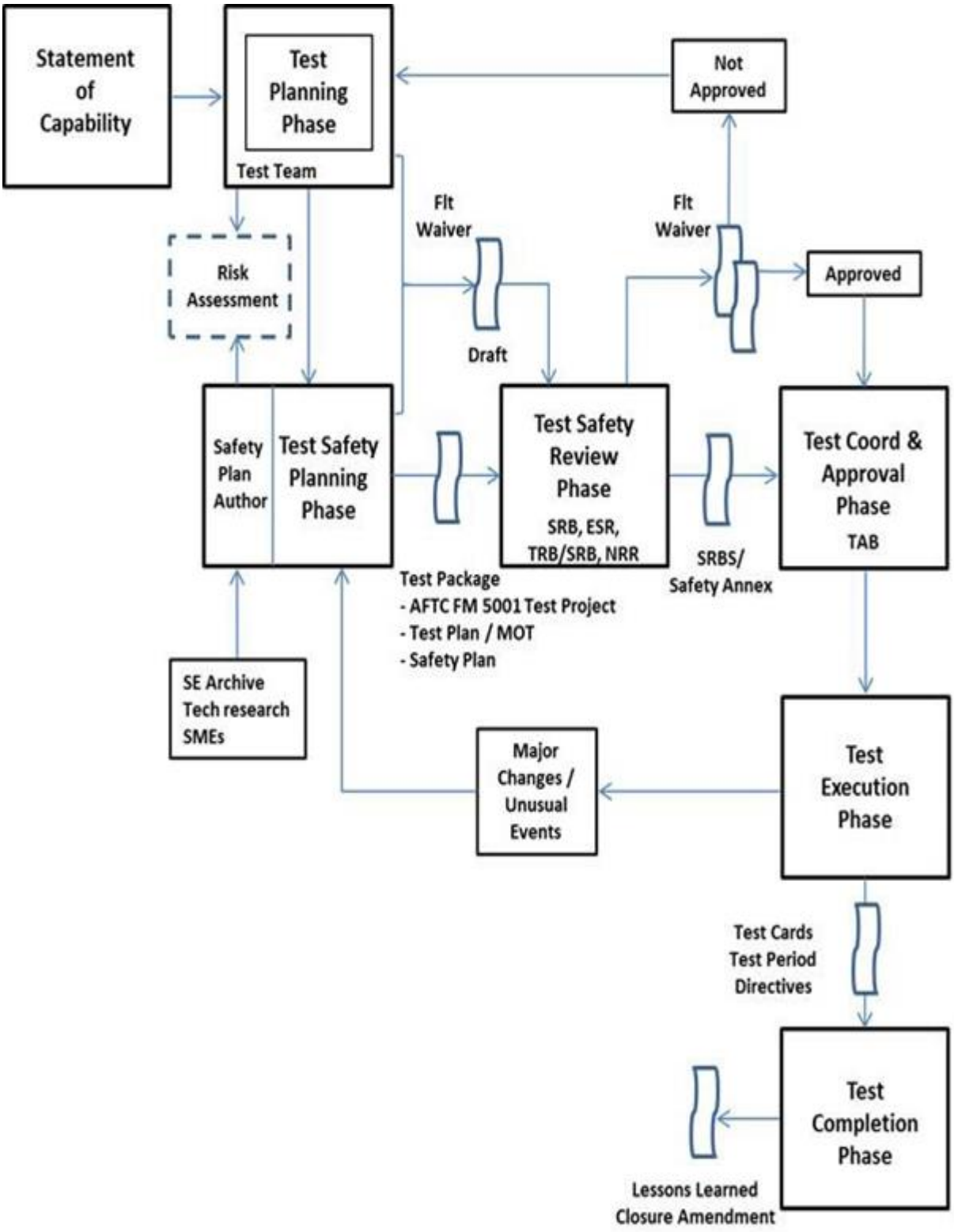
Chapter 1

INTRODUCTION

1.1. General. This instruction establishes a framework and basic requirements for AFTC test safety programs. This instruction further establishes basic vocabulary and definitions to be used universally throughout AFTC. Within the framework of this instruction, wings or their equivalent are expected to develop processes to fulfill the requirements of this instruction.

1.2. Test Safety Review Process. A Test Safety Review Process typically comprises the following functions or phases: Planning (**Chapter 3**), Risk Assessment (**Chapter 4**), Review (**Chapter 5**), Coordination and Approval (**Chapter 6**), Execution (**Chapter 7**), Revisions (**Chapter 8**), Feedback, and Test Completion and Termination. This instruction provides overall policy and guidance for test safety activity to ensure standardization of AFTC organizations while adhering to Air Force Instructions and Air Force Materiel Command Supplements. Organizations within AFTC will supplement this instruction to provide further test safety process details that uniquely apply to their specific test safety requirements. **Figure 1.1** AFTCI 91-202 Process Flow shows the phases and the typical products from each phase.

Figure 1.1. AFTCI 91-202 Process Flow.



1.2.1. As part of the review process, the units will ensure that the appropriate safety plan writers, reviewers and approvers have signed the safety planning documents during the safety review process. This can be done via a locally generated form, workflow process or other electronic review. For the Test Safety Planning Phase, these include the safety plan author, Test Safety Officer (TSO), project pilot or project test engineer, and a test unit senior-level leader (e.g., squadron commander [or equivalent], squadron commander deputy, squadron chief engineer) indicating the test unit's judgment that the safety plan is ready for the Test Safety Review Phase.

1.2.2. Additionally, during the Test Safety Review Phase the locally developed process will include a method for capturing the Safety Review Board (SRB) members' signatures, to include the SRB chair, independent operations reviewer, technical experts and any additional safety reviewers (see **paragraph 2.3**). These signatures are required before the coordination and approval phase is accomplished. The SRB chair may elect to fulfill this requirement by coordinating the final safety plan with all other SRB members for their agreement with its content and thus the SRB chair's signature represents all SRB members. As the final step in the test safety process, coordination and approval signatures must be captured. These must include the Test Execution Authority (TEA), but can also include the Wing/Complex Safety Office, Group Commander or equivalent, Wing/Complex Technical Director, Wing/Complex Commander, AFTC Safety Office and AFTC Commander. Approval level is specified in **Table 6.1**

1.3. Safety Review Process Goals. The goal of any test safety review process is to prevent mishaps during test activities. This process should identify test hazards and establish both procedures and corrective actions to eliminate or control the hazards. The process will allow independent reviewers to evaluate test unique hazards identified by the test team, assess proposed mitigations and corrective actions, and affirm or modify the test team's proposed overall risk level. Once the independent review board has agreed upon and proposed an overall risk level, the safety plan is reviewed and approved by leadership at a level appropriate for the assessed risk. Risk management must be integrated and documented into all stages of Test and Evaluation (T&E) activities to identify test hazards, control measures and acceptance/rejection of the residual risk by an appropriate TEA. The safety plan records due diligence in risk management, acceptance and communicates (provides a written copy of) hazards and mitigating measures to test personnel.

1.4. Risk Management

1.4.1. Risk Management (RM) is the systematic application of management, engineering principles, criteria and tools to optimize all aspects of safety within the constraints of mission/activity effectiveness, time, and cost throughout all mission/activity phases. RM is the main tool used to prevent mishaps and is the essence of any test safety review process within AFTC. While each test may be unique, the test safety review process for each test will follow a predictable, consistent process. The policy outlined in this instruction and the processes defined in local supplements are tailored to manage risk unique to test activity. Detailed processes for risk management can be found in AFPAM 90-803, *Risk Management (RM) Guidelines and Tools.*

1.4.2. At the discretion of subordinate units, the policy defined in this instruction and local supplement may be used to complete and approve an RM review of non-test activities.

1.5. Safety Mindset. While test safety processes should be intentionally thorough, no process is perfect. Everyone involved in test must maintain a safety mindset. A safety mindset does not assume that a test is safe simply because the test has been reviewed and approved; rather, it is continually on the lookout for previously unrecognized hazards during test planning and execution. Once recognized, appropriate actions must be taken to prevent those hazards from becoming mishaps.

1.6. Scope. This instruction applies to:

1.6.1. Any ground or flight test activity utilizing AFTC assets. AFTC assets include:

1.6.1.1. Resources owned or possessed by AFTC (personnel, aircraft, equipment, facilities, etc.).

1.6.1.2. Ranges or airspace owned or restricted for use by AFTC units.

1.6.2. Any activity where the AFTC commander or subordinate commander has responsibility for the safety of the general public such as the Major Range and Test Facility Base Commander IAW DoD 3200.11.

1.6.3. Any activity utilizing AFTC assets that presents unique hazards not covered by published procedures or management directives.

1.6.4. AFMC assets when AFTC units are assigned as Lead Developmental Test Organization (LDTO).

1.6.5. Any AFTC unit assigned or acting in the capacity of an Executing Test Organization (ETO) that is responsible for the safe conduct of test, even when AFTC assets are not at risk.

1.7. Waivers to This Instruction. The AFTC Commander is the waiver authority for this instruction. Guidance in AFI 91-202, AFMC Sup Chapter 13 would still apply unless waived separately. The AFTC Chief of Safety (AFTC/SE) may approve minor variations from this instruction provided that the intent of the test safety process and this instruction are adequately met. Any variations or waivers to this instruction that have been approved by AFTC will be on file with the Wing/Complex Test Safety Office. Waiver requests must be coordinated through the appropriate wing safety office prior to submission to AFTC/SE. For minor variations, AFTC/SE will reply with an email with concur or non-concur and a tracking number. Waivers to local supplements will be handled in accordance with the established instructions in the supplement.

1.7.1. **(Added-96TWSUP)** The waiver authority of this supplement resides with 96 TW/CC. Waivers to this supplement must be coordinated through 96 TW/SE prior to submission to the 96 TW/CC for approval; 96 TW/SE will provide a recommendation for approval or denial to 96 TW/CC with the waiver request. All approved waivers to this supplement must be kept on file with the organization requesting the waiver and included in the Test Directive for which the waiver is requested or applicable.

1.8. Authority. Compliance with AFTC Test Safety Review Policy does not provide authority to violate Air Force, AFMC, or AFTC instructions or directives or flight manual guidance.

1.8.1. When a test activity must deviate from an AFI or other command directive, units will comply with the applicable waivers/deviations process outlined in the applicable document. A copy of the waiver will be filed with the safety office and/or test unit. If the waiver authority is within the local Wing/Complex chain of command, the waiver may be obtained during the approval cycle and documented as a coordination comment within the safety plan.

1.8.1.1. **(Added-96TWSUP)** Any waivers to existing instruction or command directive must be presented to the Safety Review Board (SRB). The SRB will provide a risk assessment based on waiver approval. If the waiver is ultimately denied, the SRB will reconvene and reassess the risk.

1.8.2. When a test activity must deviate from a technical order (T.O.) or flight manual, units will follow current command guidance (AFI 11-215, *USAF Flight Manuals Program (FMP)* and AFI 11-215_AFMCSUP1, *USAF Flight Manuals Program [FMP])* or program office guidance for uninstalled test items. If a waiver is required, a copy of the draft waiver will be included in the safety plan for discussion by the SRB. Test teams will note the deviation in the test plan and incorporate safety planning as required during the risk assessment process. The approved waiver must be included in the test package.

Chapter 2

SAFETY RESPONSIBILITIES

2.1. Test Safety Approval and Coordination Responsibilities

2.1.1. Responsibilities of personnel/organizations involved in the test safety approval and coordination phase are as follows:

2.1.2. The AFTC Commander will:

2.1.2.1. Be the approval authority for this instruction.

2.1.2.2. Be the waiver authority for this instruction.

2.1.3. AFTC/SE will:

2.1.3.1. Establish test safety review policy for all AFTC organizations.

2.1.3.2. Review local supplements to this instruction.

2.1.3.3. Approve minor variations from this instruction that meet the intent of the test safety process and this instruction.

2.1.4. AFTC Test Safety Office will:

2.1.4.1. Conduct an annual test safety process review with all AFTC organizations to review and refine test safety best practices.

2.1.4.2. Assess compliance of AFTC organizations with this instruction and appropriate Management Internal Control Toolset (MICT) Communicators during site visits, staff assisted visits (SAVs) and virtually through MICT.

2.1.4.3. Approve locally developed training in support of Test/System and locally developed supplements to this AFTCI.

2.1.4.4. Notify HQ AFMC/SE, AFMC/A3 and asset owner of high risk tests, IAW AFI 91- 202_AFMCSUP.

2.1.5. Wing/Complex Safety Office will: Approve minor variations from the supplements to this instruction, provided that the intent of the test safety process and this instruction are adequately met.

2.1.6. Wing/Complex Test Safety Office (or SE delegate if none exists) will:

2.1.6.1. Develop a local test safety review process as a supplement to this instruction.

2.1.6.2. Maintain the integrity of locally developed test safety review process to ensure independent government review of safety planning documentation is being accomplished for leadership approval decisions.

2.1.6.3. Develop and maintain test safety training programs. Provide initial test safety review process training for Wing/Complex/Unit personnel (including contractor personnel as appropriate) who are involved in test safety planning, review, coordination and/or approval, to include independent safety reviewers/subject matter experts as described in Section 2.3. Annual training will be provided for safety plan authors. Training products will be updated when this instruction, or supplements to this instruction, is revised.

2.1.6.4. Incorporate lessons learned and best practices into appropriate training programs and provide for discussion during AFTC's annual test safety process review.

2.1.6.5. Provide guidance and assistance to safety plan authors on test safety planning.

2.1.6.6. Designate or act as the SRB chairperson.

2.1.6.7. Ensure an archive of approved safety plans and associated documentation is current, maintained in a searchable archive or electronic folder and available to test teams across the enterprise.

2.1.6.8. IAW AFI 91-202, AFMCSUP, paragraph 13.10, lessons learned, effectiveness of hazard controls or minimizing procedures, unexpected hazards, value added from the safety review process, and suggestions for improving the safety review process will be captured in a lessons learned archive and available across the enterprise in a searchable format. This can be done at the completion of the program or at program reviews as a joint effort between the test team and the test safety office.

2.1.6.9. Develop and maintain a cadre of test safety officers to support the test safety process. The test safety officers may be part of the test unit or in the test safety office. If part of the test unit, they will be designated in writing by the test unit.

2.1.6.10. Test Safety Officers will:

2.1.6.10.1. Assist test teams with identification of test hazards and appropriate mitigation measure and preparation of all safety-related documentation, including amendments from safety planning through the approval phases.

2.1.6.10.2. Sign the AFTC Form 5001, *Test Project Safety Review*, or equivalent, which shows that the safety-related documentation complies with content and format standards contained in this instruction and supplements to this instruction.

2.1.6.10.2.1. **(Added-96TWSUP)** The 96 TW Test Directive has been approved as an equivalent document indicating the completion of the Test Safety Review Process for 96 TW activities. Appropriate signatures on the Test Directive, Method of Test Appendix and the Safety Appendix will be identified as water marks placed by the electronic workflow for these documents. The Test Execution Authority signature on the TD signifies risk acceptance and test approval.

2.1.6.10.3. Complete the appropriate Wing/Complex test safety training course.

2.1.6.10.4. Advise test team and safety plan author on appropriate independent reviewers and subject matter experts available for test and safety plan development.

2.1.7. **(Added-96TWSUP)** The 96 TW Test and Range Safety Office (96 TW/SEU) is the OPR for this supplement and responsible for the administration and implementation of the AFTC Test Safety Review Policy for all 96 TW test and training operations. Responsibilities assigned to the Wing/Complex Test Safety Office and verbiage referencing the Wing/Complex Test Safety Officer refer to 96 TW/SEU. Any exception will be as directed by the 96 TW Chief of Test and Range Safety.

2.1.7.1. **(Added-96TWSUP)** Test Safety Officers supporting 96 TW test activities are assigned to the Test and Range Safety Office, 96 TW/SEU. The Chief of Test and Range Safety will provide a Test and Range Safety Engineer (T/RSE) who will:

2.1.7.1.1. **(Added-96TWSUP)** Complete the Test and Range Safety Officer certification training.

2.1.7.1.2. **(Added-96TWSUP)** Serve as the Test and Range Safety Officer for the program/activity.

2.1.7.1.3. **(Added-96TWSUP)** Chair the Safety Review Board (SRB).

2.1.7.1.4. **(Added-96TWSUP)** Publish the final safety plan in the TD's Safety Appendix,

2.1.7.1.5. **(Added-96TWSUP)** Document the minutes of the SRB, approve any submitted Test Hazard Analysis (THA) forms based on the SRB discussion and prepare the Safety Appendix for 96 TW/SE approval.

2.1.8. **(Added-96TWSUP)** For Air Force munitions, the Chief of Systems Safety (96 TW/SES) will provide a Systems Safety Engineer who will:

2.1.8.1. **(Added-96TWSUP)** Be a member of the SRB and be knowledgeable on the system(s) or munitions that are being tested or used in training.

2.1.8.2. **(Added-96TWSUP)** Provide a design safety certification statement to the SRB for the weapon system(s) and/or munitions being tested or used for training.

2.1.9. **(Added-96TWSUP)** For activities involving any ammunition, flares, explosive devices, aircraft store/weapon systems or directed energy weapon, the Chief of Weapons Safety (96 TW/SEW) will provide a representative to the SRB.

2.2. Test Unit Safety Planning Responsibilities

2.2.1. Responsibilities of personnel within a test unit (e.g., a squadron or Combined Test Force) during the test safety planning and review phase are as follows:

2.2.2. Squadron Commanders (Test Unit Commander, Director or equivalent) will:

2.2.2.1. Review and provide coordination for all test and safety plans within their organization where they are not the ETO.

2.2.2.2. Ensure all unit personnel involved in safety planning or execution are familiar and comply with this instruction and local supplements and receive test safety training.

2.2.2.3. Support the AFTC test safety process, which may include operations and/or technical personnel assigned to their test unit participating in independent review of other test projects or activities.

2.2.2.4. Provide Test Safety Officers (TSOs), as applicable per Wing/Complex test safety policy.

2.2.2.5. Maintain a list of test safety officers with training and experience applicable for test unit projects.

2.2.3. Safety plan authors will:

2.2.3.1. Complete a locally developed Test/System Safety training course offered by the Wing/Complex Test Safety Office and approved by AFTC/SET.

2.2.3.2. Maintain currency by completing continuation training annually.

2.2.3.3. Develop safety plans in accordance with **Chapter 3** of this Instruction and local supplements.

2.2.3.4. Review and sign the applicable AFTC Form 6239, *T-2 Modification Airworthiness Compliance*, for aircraft modifications that have an airworthiness assessment of *Impact* to ensure the hazards identified in the form that affect test safety are addressed in the test safety planning. Attendance at the Design Review Board (DRB)/Configuration Control Board (CCB) is highly recommended for upfront and early insight into the modifications and airworthiness assessments.

2.2.3.5. Ensure safety plans clearly and adequately provide enough information to support an approval decision.

2.2.3.6. Identify a proposed project risk and include the rationale for the proposed risk level to the safety review board members in support of the safety reviews described in section 5.2.

2.2.3.7. Ensure safety plans include a summary of lessons learned and Test Hazard Analyses (THAs) from similar and/or applicable tests to indicate if there are any applicable hazards to consider in approval of the Safety Plan.

2.2.3.8. Nominate the team of independent safety reviewers to the SRB chair for approval. The SRB chair can also require other independent safety reviewers.

2.2.3.8.1. **(Added-96TWSUP)** A list of approved Independent Safety Reviewers (ISR) for 96 TW SRBs is maintained by 96 TW/SEU. The safety plan author may nominate the ISR when submitting the required documentation.

2.2.3.9. **(Added-96TWSUP)** 96 TW Test Engineers (TE)/Program Engineers (PE) (or equivalent) are considered the safety plan authors as referenced in AFTCI 91-202. TEs will submit an initial safety plan along with the TD Method of Test (MOT) or Concept of Operations (CONOPS) to 96 TW/SEU to initiate the Test Safety Review Process as well as other required documents identified in **Table 5.1**

2.2.4. Test Team will:

2.2.4.1. Determine if test methods, conditions, and resources in test methodology balance safety and data needs.

2.2.4.2. Ensure all appropriate test techniques were considered. Choose the lowest risk technique which efficiently meets test/data objectives.

2.2.4.3. Ensure appropriate test unique hazards related to test methods and system(s) operation are identified and sufficiently controlled (eliminated, mitigated, or residual risk believed to be acceptable).

2.2.4.4. Ensure tests are being conducted per published technical orders and Air Force Instruction guidance.

2.2.4.5. Ensure flight manual waivers are submitted and approved per AFI 11-215, or per program office guidance for uninstalled test items.

2.2.4.6. Report changes to the Test Directive/Method of Test (MOT)/Test Plan to the Wing/Complex test safety office per local guidance.

2.2.4.6.1. **(Added-96TWSUP)** Document any changes to the objectives, resources and/or configurations, and procedures in an amendment to the TD MOT or CONOPS and report these changes to the SRB Chair for review and evaluation of the effect on the risk assessment.

2.2.4.6.2. **(Added-96TWSUP)** The SRB Chair will determine the appropriate course of action based on the documented changes.

2.2.4.7. Perform a review of the safety plans for their test projects every three years IAW **paragraph 8.4**

2.2.4.7.1. **(Added-96TWSUP)** The periodic review will be coordinated with the Test and Range Safety Office, specifically the T/RSE supporting the program. The T/RSE will provide documentation supporting the review and indicate the validity of the risk assessment and any issued safety requirements.

2.2.5. **(Added-96TWSUP)** The 96 TW or other responsible or participating test organization will provide a TE, Test Manager (TM) or equivalent who will:

2.2.5.1. **(Added-96TWSUP)** Participate in the SRB for each of their programs. The SRB may be chaired by the 96 TW Test and Range Safety Office or by the cognizant safety organization at the specific location.

2.2.5.2. **(Added-96TWSUP)** Ensure personnel involved in the test are briefed on the hazards, minimizing procedures and the requirements stipulated in the SA to the TD or other published safety document(s).

2.2.6. **(Added-96TWSUP)** For airborne tests, the 96th Operations Group (96 OG) or other test agency using the 96 TW facilities will assign a primary aircrew member who will:

2.2.6.1. **(Added-96TWSUP)** Assist the TE in preparation for the SRB to include participating in identifying any test unique hazards, proposing any hazard minimizing procedures and completing AFTC Form 5000 to be delivered prior to the SRB.

2.2.6.2. **(Added-96TWSUP)** Attend and participate in the SRB.

2.2.6.3. **(Added-96TWSUP)** Ensure that applicable hazards and minimizing procedures identified in the SA and applicable TEA stipulations are included in the aircrew test/run cards and are addressed during the aircrew pre-mission brief.

2.3. Independent Safety Reviewer Responsibilities

2.3.1. Independent Safety Reviewers (ISRs) include the technical experts, operations reviewers and the SRB chair. The ISRs should be from outside the test unit, which is especially important for tests proposed by the test team to be Medium or High Risk, but they can be from within the test unit. In both cases, the ISRs must not have a vested interest in the successful accomplishment of the test objectives and were not directly responsible for the development of the safety plan. The ISRs must be independent of the test project (e.g., not a project engineer or project pilot for the test), not have been involved (or had limited involvement) in preparing the test plan (MOT) or safety plan, and not the TEA. The ISRs should have appropriate qualifications. They should be senior in test experience or have formal Test Pilot School training, have applicable knowledge and sufficient expertise in the test activity to be reviewed. To the maximum extent possible, independent safety reviewers should be the same individuals that served as independent reviewers for the technical review (if applicable). Independent reviewers will be approved by the SRB chair in accordance with qualification guidelines set forth in local supplements to this instruction. All independent reviewers must have accomplished the initial test safety review process training. Independent reviewers will review and approve all THAs, Baseline Hazard Analyses (BHAs) and General Minimizing Procedures (GMPs) as part of the SRB process. Individual reviewer responsibilities are as follows:

2.3.1.1. **(Added-96TWSUP)** The 96 TW/SEU T/RSE assigned to a particular test or training program will provide the necessary safety support throughout the program. The T/RSE provides safety guidance for specific operations conducted at the Eglin Test and Training Complex (ETTC). The 96 TW Safety Office maintains its independence by having no vested interest in the cost, schedule or successful outcome of the test/training program being supported; their only interest is in the safe conduct of the test/training operation.

2.3.1.2. **(Added-96TWSUP)** The 96 TW executes ground, weapon, and flight test and training programs. Independent Reviewer qualifications will be based upon the type of program requiring safety review. 96 TW/SEU will determine if an individual is qualified to participate in an SRB as an Independent Reviewer. Risk assessment inputs by approved Independent Reviewers will be without a regard to cost, schedule and successful outcome of the test/training program.

2.3.1.3. **(Added-96TWSUP)** Every effort will be made to include local independent and expert participants in the conduct of 96 TW SRBs. The SRB chairperson will determine if those individuals are qualified to fill the roles of technical and operational reviewers, have the required training and will designate board members as Independent Reviewers. If a local Independent Reviewer is not available, the SRB chairperson will ensure Independent Reviewers can participate in some fashion during the scheduled SRB. The test team should work with the T/RSE well in advance of the planned SRB in order to obtain a non-local Independent Reviewer.

2.3.1.4. **(Added-96TWSUP)** Independent Safety Reviewers (ISR) will also complete the 96 TW ISR Risk Assessment training available through the 96 TW Test and Range Safety Office. ISR will:

2.3.1.4.1. **(Added-96TWSUP)** Participate in the SRB.

2.3.1.4.2. **(Added-96TWSUP)** Coordinate on the Safety Appendix detailing the risk assessment associated with the test or training operation as documented in the TD. The coordination on the Safety Appendix will include providing a concurrence or non-concurrence of the Safety Appendix. If a non-concurrence is provided, then justification will be required from the ISR and documented in the Safety Review Board Summary (SRBS) for notification to the TEA. The TEA will consider this as part of the approval process.

2.3.2. SRB Chair will:

2.3.2.1. Ensure appropriate test unique hazards and routine hazards that can be exacerbated by the test conditions are identified and sufficiently controlled (eliminated, mitigated, or residual risk believed to be acceptable).

2.3.2.2. Ensure general and special mitigation measures are clear and unambiguous.

2.3.2.3. Ensure the safety assessment is clearly and concisely articulated to approval authorities.

2.3.2.4. Ensure operations, facilities, maintenance, etc., reviewers have appropriate expertise relevant to the type of testing being reviewed.

2.3.2.5. Not be under the control or influence of the organization responsible for operations and execution of the test and have direct lines of communication with the commander.

2.3.2.6. Be a government employee.

2.3.2.7. Be a full-time safety staff or formally designated and approved by the Wing/Complex Chief of Safety.

2.3.2.8. Determine if another safety review composed of a different set of ISRs is required when the SRB includes ISRs from within the test unit and the SRB risk assessment is higher than the test team's proposed risk level.

2.3.3. Technical Reviewer will:

2.3.3.1. Ensure safety hazards are identified and appropriately controlled (eliminated, mitigated, or residual risk believed to be acceptable).

2.3.3.2. Have applicable knowledge and sufficient expertise in the test activity to be reviewed.

2.3.3.3. Be independent of the test team.

2.3.4. Operations Reviewer will:

2.3.4.1. Be experienced in the type of system under test (SUT) such as aircraft (i.e., fighter, bomber, cargo), ground test facility (i.e., wind tunnel, sled track, propulsion stand, climatic lab), and the types of tests being conducted. Exceptions can be approved by the Wing/Complex Chief of Test Safety.

2.3.4.2. Ensure tests are executable, all test techniques were considered, and lowest risk technique which efficiently meets test/data objectives was selected.

2.3.4.3. Ensure hazards related to operating the system are identified and appropriately controlled (eliminated, mitigated, or residual risk believed to be acceptable).

2.3.4.4. Be independent of the test team.

2.3.5. Facility Reviewer (if required) will: Ensure hazards related to operating ground test facilities are identified and appropriately controlled.

2.3.6. Maintenance Reviewer (if required) will: Ensure test conduct and execution does not deviate from SUT maintenance procedures or technical manuals.

2.3.7. Range Safety/Range Operations Engineer (if required) will: Analyze proposed test plans and attend SRBs relating to range activities as deemed necessary by the SRB chair.

2.3.8. Flight Safety representative (if required) will: Establish procedures to coordinate on all planned or contractual flight tests.

2.3.9. Weapons Safety representative (if required) will: Review weapon safety analyses, operating instructions, and attend SRBs relating to aircraft store/weapons system and range activities where new or modified weapon testing or explosives is involved.

2.3.10. In any SRB, additional expertise from other sources may and should be called upon when required. Optional Reviewers, as deemed necessary by the SRB chair, may include, but are not limited to:

2.3.10.1. Test Engineer

2.3.10.2. System Safety Engineer

2.3.10.3. Occupational Safety Representative

2.3.10.4. Explosive Ordnance Disposal Representative

2.3.10.5. Test Requestor / Item Contractor

2.3.10.6. Airspace Representative

2.3.10.7. Logistics Representative

2.3.10.8. Munitions Representative

2.3.10.9. Fire Department Representative

2.3.10.10. Bioenvironmental Engineer

2.3.10.11. Medical Representative

2.3.10.12. Environmental Management Office Representative

2.3.10.13. Range O&M Representative

2.3.10.14. Laser or Directed Energy Safety Representative

2.3.10.15. Flight Termination System Analyst

Chapter 3

TEST SAFETY PLANNING PHASE

3.1. Test and Safety Planning. Safety planning and test planning are integral and iterative processes, and as such, both should be interwoven to ensure the test methods incorporate safety controls where possible. Well planned tests that consider and incorporate risk control measures to eliminate or mitigate test hazards are inherently safer than test plans without this safety emphasis. This chapter covers considerations and guidance during the test safety planning and review phases.

3.2. Safety Considerations during Test Planning

3.2.1. Test Approach or Build-up. During test plan development, the test team will carefully consider the test approach or build-up. The way the test approaches a hazardous or unknown condition must be clearly defined. If predictive analysis does not exist, or has questionable validity, the test methodology may require a more refined buildup approach to offset the risk. Criteria to continue, or more importantly when to stop, can provide good risk control by providing a clearly defined roadmap into the test team's decision making. This decision-making process is extremely important and should be documented.

3.2.2. Test Plan Size and Complexity. The test team must consider the size and complexity of the test plan and assess whether a review of a large, complex safety plan is more or less advantageous than several smaller reviews. If feasible, teams may conduct test safety planning for large, complex test plans in smaller, less complex safety plans matched to progressive phases of the test project.

3.2.3. Integration. If the planned testing utilizes more than one test plan, method of test, test information sheet (TIS), or procedure, it is incumbent upon the team to provide a clear test progression description. Without a clear path, the ability to identify hazards appropriately and develop a sensible risk assessment is difficult. The test team should be aware of this basic issue to avoid significant and unplanned schedule delays caused by action items or cancelled safety review boards.

3.3. Safety Planning Objectives. The objective of the safety planning phase is to identify and assess hazards and develop controls or mitigation measures to reduce the risk to an acceptable level.

3.3.1. Hazard Identification. The first step in safety planning is identification and evaluation of existing and potential hazards. Some hazards will be inherent in operating the system and others will be induced by the test itself. For test safety planning, the goal is to identify and mitigate test unique hazards. If the nature of the test increases the probability or severity of non-unique hazards they should be addressed, mitigated, and documented.

3.3.1.1. Identify Test Unique Hazards. The team will identify unique hazards associated with each type of test or activity. In some cases test activities may elevate the risk associated with routine operational hazards, thus requiring additional safety planning. In the safety documentation, hazards should adequately describe the risk situation including the unsafe act or condition and its effects. It is often helpful to assess the risk (probability and severity, see **Chapter 4**) prior to applying mitigations as well as after mitigations are in place. Sources for identifying test unique hazards include:

3.3.1.1.1. Archived test and safety planning, to include lessons learned and THAs, across the enterprise for consideration of similar tests.

3.3.1.1.2. Personnel or test teams with experience in similar test activities or testing.

3.3.1.1.3. Technical libraries, internet, etc. to research technical aspects.

3.3.1.1.4. System safety hazard analyses of the test article and test facility.

3.3.1.1.5. Applicable safety reviews from other organizations such as the Program Office, Nonnuclear Munitions Safety Board, Directed Energy Safety Board, 711 Human Performance Wing Independent Review Board or the contractor.

3.3.1.1.6. Aircraft modification documents.

3.3.1.2. **(Added-96TWSUP)** A few typically generic hazards that may become test unique due to the nature of the test should also be considered.

3.3.1.2.1. **(Added-96TWSUP)** Bird strikes are not generally considered to be a test-unique hazard. But if, during the execution of a test, flight through a known bird hazard area is necessary, particularly during migration seasons, through migration routes and during peak migration time of day/night then “bird strikes” would be a test-unique hazard.

3.3.1.2.2. **(Added-96TWSUP)** Heavy equipment operations are not generally considered to be a test-unique hazard. But if during the execution of a test, a novel aircraft loading method is proposed using non-standard heavy equipment, then “heavy equipment operation” would be a test-unique hazard.

3.3.1.2.3. **(Added-96TWSUP)** Mid-air collision is not generally considered to be a test-unique hazard. But if during the execution of a test, safety/photo chase requirements could potentially increase the likelihood of proximity or relative translation at periods when the aircrew could be distracted while executing other tasks, then “chase aircraft collision” would be a test-unique hazard.

3.3.1.2.4. **(Added-96TWSUP)** Additional non-test unique hazards identified should also be documented in the proposed safety plan and discussed at the SRB.

3.3.1.3. **(Added-96TWSUP)** It may be difficult to tell when a standard parameter or action becomes test-specific. Some good indications are:

3.3.1.3.1. **(Added-96TWSUP)** If the test includes events that could be characterized as a system demonstration, system variant, or system combination for the first time.

3.3.1.3.2. **(Added-96TWSUP)** If the test includes demonstrating systems at conditions outside the bounds of previous demonstrations.

3.3.1.3.3. **(Added-96TWSUP)** If the test involves significantly modifying standard procedures.

3.3.1.3.4. **(Added-96TWSUP)** If personnel unfamiliar with the test were to conduct the test and needed or wanted a briefing to better understand the associated hazards.

3.3.1.3.5. **(Added-96TWSUP)** If a nominal hazard probability is increased because of the nature of the test.

3.3.2. Eliminate or Control Hazards. Once the causes of each hazard have been identified, minimizing procedures or controls are used to reduce risk by reducing severity or probability or both. The following order of precedence should be applied to eliminate or control any hazards identified during the safety planning.

3.3.2.1. Design the test to eliminate the probability of the hazard occurring. This could include a decision to not perform the test if the risk is deemed to be unacceptably high. A redesign of the system to eliminate the hazard is another option.

3.3.2.2. Change the test methodology to reduce the probability, severity, or exposure to the hazard (building up to the test condition can be a strong control method).

3.3.2.3. Incorporate safety devices (e.g., spin chute or additional power sources).

3.3.2.4. Provide caution and warning devices to detect an unsafe condition or trend or install instrumentation and data displays with active monitoring.

3.3.2.5. Develop procedures and training when it is impractical to change the design or test methodology.

3.4. Test Package Documentation

3.4.1. The “test package” shall be an all-encompassing package of documents consisting of a test plan, safety plan, and any other appendices or documentation that support the test planning. All safety planning will be documented in the safety plan including minimizing procedures, THAs, and safety buildup that may be in the test planning documents. Additional guidance on the test planning process and documentation can be found in local Wing/Complex test planning instructions.

3.4.1.1. **(Added-96TWSUP)** The official 96 TW test package is known as the Test or Training Directive. Appendices to the TD may include the MOT and/or CONOPS, Technical Support Appendix (TSA), Logistics Appendix (LA) and Safety Appendix. For the purposes of the implementation of AFTCI 91-202, the 96 TW Test Directive’s MOT/CONOPS and Safety Appendix serves as the “test package.”

3.4.1.1.1. **(Added-96TWSUP)** The MOT details the test request, objectives, test items, ground and/or airborne support, test procedures or CONOPS and potential hazards. Proposed minimizing procedures to identified hazards can be integrated in the MOT and will be reviewed as part of the SRB briefing (**AFTCI 91-202, Chapter 5**). These procedures can also be found documented in the THA form or detailed in a separate MOT/CONOPS potential hazards paragraph; if the safety plan is documented in the MOT/CONOPS, verbiage indicating "The Safety Appendix is the final authority for the identified hazards, the required minimizing procedures and associated risk assessment" will be included. The safety plan procedures are the focal point of the SRB brief. The approved details/resources of the test are documented in the Safety Appendix (**Attachment 2** provides an example of the Safety Appendix format).

3.4.1.1.2. **(Added-96TWSUP)** The safety plan is a test organization's plan identifying any unique hazards and proposed mitigations. The initial safety plan submitted by the Test Engineer will be discussed and modified as necessary through the SRB and finalized in the published Safety Appendix.

3.4.1.1.3. **(Added-96TWSUP)** The THA form, as a stand-alone working-level document, is not intended to be a directive document requiring the procedures defined on the form to be followed; those requirements will be defined in the published and signed Safety Appendix. The THA form communicates the logic behind the risk assessment for test-unique hazards and provides reference safety material for future test teams who may conduct similar tests or encounter similar hazards; instruction on filling out the THA form is included in **Attachment 3**. Members invited to the SRB will ensure THA forms are completed and delivered to the SRB Chairperson prior to the SRB. The SRB will discuss and approve the proposed risk level and minimizing procedures or assign a new risk level and/or modify the proposed minimizing procedures. The risk assessment will be based on the finalized proposal taking into account the procedures whether modified or otherwise. The THA will be documented in the SRBS or in the THA form as an attachment to the SRBS based on the discussion and outcome of the SRB. A database of previously issued THA forms is maintained for the test community to search past hazards and their minimizing procedures.

3.4.2. The Safety Plan should follow documentation guidance from Chapter 13, paragraph 13.6.5, of AFI 91-202, *The US Air Force Mishap Prevention Program* as supplemented by AFMC. The Safety Plan, at a minimum, is comprised of THAs, GMPs, BHAs, a Baseline Safety Review (if applicable), the Safety Review Board Summary, mishap accountability, and waivers/deviations from AFIs, T.O.s and flight manuals. Format and structure of the safety plan may be further defined in local supplements to this instruction. In the event testing requires the preplanned damage/destruction of test assets, AFI91-202, AFMC Supplement paragraph 13.11.6 should be followed.

3.4.2.1. THAs are used to document and identify test hazards and the actions necessary to minimize or control them. Each THA captures a test unique hazard. A hazard is any condition that has the potential of causing a mishap. Confirm that the hazard is not a hazard associated with the basic operation of the aircraft, vehicle, SUT, or facility. If the hazard is not unique to the series of tests, no THA is required. For example, midair collision with non-participating aircraft and bird strikes are not generally considered test unique hazards. However, should the very nature of the test increase the probability of these hazards above that of normal operations, they should be addressed as test unique hazards. THAs will be documented on a *Test Hazard Analysis* document that captures the information required in AFI 91-202, AFMC supplement. The THA will include the following:

3.4.2.1.1. Mishap severity and probability of the hazard as discussed in detail in **Chapter 4**.

3.4.2.1.2. Causes are anything that could lead to the presence of the hazard. The causes can include inherent hazardous characteristics, design inadequacies, hardware failures, environmental effects, software deficiencies or operator errors. This is the cause of the hazard, not the mishap. There may be more than one cause for each hazard.

3.4.2.1.3. Effect is the outcome if the hazard is not controlled. The effect is what the THA is trying to prevent and is directly related to the mishap severity level. Effects are often descriptors that tie into the mishap severity, such as loss of life/aircraft/facility, severe injury/damage, minor injury/damage, superficial injury or less than minor damage.

3.4.2.1.4. Controls or Minimizing Procedures should be an action or procedure and tied to a specific cause, causes, or effect it is trying to control. These controls or minimizing procedures attempt to break the chain of events linking the causes to the hazard.

3.4.2.1.5. Corrective Actions or Emergency Procedures are the list of actions taken to prevent or mitigate a mishap (the effect) if the hazard occurs. Actions may be taken by the control room personnel, ground personnel, flight crew, test facility operators, and anyone else participating in the test. Test unique and hazard specific emergency procedures would be listed here. If not test unique, corrective actions may state that operation manual procedures will be followed. These corrective actions attempt to break the chain of events linking the hazard to the mishap.

3.4.2.1.6. Comments are optional information that helps support the THA risk analysis but are not directive in nature and do not contribute to breaking the mishap chain.

3.4.2.1.7. While hazard identification should have been accomplished leading up to the SRB, emphasis should be placed on identifying items of special interest for THA/SRB consideration including, but not limited to:

3.4.2.1.7.1. New systems or system variants: aircraft, stores, instrumentation, test equipment.

3.4.2.1.7.2. Unique and/or unprecedented systems not previously used in the test environment: aircraft, stores, instrumentation, test equipment.

3.4.2.1.7.3. **(Added-96TWSUP)** Additional considerations should also be given to:

3.4.2.1.7.3.1. **(Added-96TWSUP)** Expansion of development systems to new user groups/test teams

3.4.2.1.7.3.2. **(Added-96TWSUP)** Unfamiliar procedures

3.4.2.1.7.3.3. **(Added-96TWSUP)** Transitions of responsibility

3.4.2.1.7.3.4. **(Added-96TWSUP)** Modifications to established procedures or test resources

3.4.2.1.7.3.5. **(Added-96TWSUP)** Unproven or seldom used test methods

3.4.2.1.7.3.6. **(Added-96TWSUP)** Unfamiliar/unproven supporting analytical techniques

3.4.2.1.7.3.7. **(Added-96TWSUP)** First-time test events for the system (e.g., aircraft, aircraft component, store) under test or for the specific system configuration, or combination of systems (e.g., aircraft load out, electronic subsystem combination), under test

3.4.2.1.7.3.8. **(Added-96TWSUP)** Recovery of test items to include fuses and/or components

3.4.2.1.7.3.9. **(Added-96TWSUP)** Testing to address changes in systems under test, whether due to:

3.4.2.1.7.3.9.1. **(Added-96TWSUP)** Pre-planned upgrades

3.4.2.1.7.3.9.2. **(Added-96TWSUP)** Modifications resulting from deficiencies identified by inspection/ review

3.4.2.1.7.3.9.3. **(Added-96TWSUP)** Corrective actions addressing failures in previous related tests

3.4.2.1.7.3.9.4. **(Added-96TWSUP)** Test approaches with a history of associated failures/problems

3.4.2.1.7.3.9.5. **(Added-96TWSUP)** Systems/subsystems/components with a history of associated failures/problems

3.4.2.1.7.3.9.6. **(Added-96TWSUP)** Additional research and/or analysis

3.4.2.1.7.3.9.7. **(Added-96TWSUP)** Test range restrictions (location, weather, additional resources)

3.4.2.1.7.3.9.8. **(Added-96TWSUP)** Test point/phase changes (reorder points, insert/complete phase, etc.)

3.4.2.1.7.3.9.9. **(Added-96TWSUP)** Test point specific requirements (additional requirements for elevated risk test points)

3.4.2.1.7.3.9.10. **(Added-96TWSUP)** Familiarization or additional training

3.4.2.1.7.3.9.11. **(Added-96TWSUP)** Aircrew restrictions (experience level, qualifications, number aboard, required equipment, etc.)

3.4.2.1.7.3.9.12. **(Added-96TWSUP)** Aircraft restrictions (capabilities or equipment that would simplify/ enhance test conduct)

3.4.2.1.7.3.9.13. **(Added-96TWSUP)** Munitions modifications

3.4.2.1.8. **(Added-96TWSUP)** Taking all entered information into consideration, the risk assessment is determined using the Risk Level Assessment and the definitions found in AFTCI 91-202, Chapter 4. This defines the overall risk level for the identified hazard. Any test or training point/event subject to this hazard would then be assessed according to highest hazard risk.

3.4.2.2. GMPs are stand-alone phrases/statements and are used to address SUT restrictions, test build-up, critical parameter monitoring, go/no-go criteria, weather or environmental criteria, and flight test chase requirements among other items of test safety concern.

3.4.2.3. Baseline Hazard Analysis (BHA): An analysis used to document known hazards concerned with the normal day-to-day operation of an aircraft, system, subsystem or facility. Examples of test equipment or test facilities which may be a good candidate for BHAs are: wind tunnels, high-speed sled track, centrifuge, anechoic chamber, or climatic laboratory. BHAs shall include supporting documentation in order to assist reviewers of the analysis. BHAs can also be used to document hazards associated with instrumentation packages. BHAs will be documented in accordance with local supplements. In addition to severity, probability, cause, effect, mitigation measures, and corrective actions or emergency procedures, the following shall be included on all BHAs:

3.4.2.3.1. Hazard analysis title. The title should be easily searchable and shall describe the system or process analyzed. The title shall include the configuration item.

3.4.2.3.2. Description. The description shall include the system, test, or process being evaluated, the purpose, major system components, energy sources, interfaces (system and human), operating location and environment. Assumptions shall be clearly stated. Revisions to an approved document shall include a summary of the revision.

3.4.2.4. Baseline Safety Report (BSR): A compilation of the entire BHAs. The BSR allows the individual hazard analyses that make up the baseline to be evaluated in a comprehensive package and thus shows the interaction of the systems and interfaces.

3.4.2.5. Safety Review Board Summary (SRBS). The SRBS documents the results of the SRB meeting, any open action items that require closure prior to the final approval of the test package, and the risk assessment. Final approval of the SRBS resides with the SRB chair. As a minimum, the SRBS will contain:

3.4.2.5.1. Date of SRB meeting.

3.4.2.5.2. SRB attendees.

3.4.2.5.3. SRB action items and responses, and coordination comments and responses.

3.4.2.5.4. Overall risk assessment with justification.

3.4.2.5.5. Any test/training activity contingent on any waivers (i.e. chase waiver, deviations from AFTCI 91-202 and/or local supplements) or flight manual waivers per AFI 11-215 requires discussion at the SRB and will be included in any hazard risk assessment and documented in the SRBS. Any waiver not approved by the TEA, or appropriate approval authority, after the SRB will require a reassessment by the SRB. The SRBS will identify how the hazard risk assessments apply to the proposed test points for tests which contain identifiable test points, test sets or test matrices.

3.4.2.6. Mishap Accountability. Detailed information on mishap accountability and investigating responsibility must be provided by the test team in the safety plan when deviating from AFI 91-204, or if multiple MAJCOMs are involved, or if non-Air Force assets are involved, to include pre- mishap planning. A memorandum of agreement is the preferred method when multiple agencies are involved. For tests that include non-AFTC resources, the AFTC assets that are at risk for the test should be explicitly identified.

3.4.2.7. Other items that should be included are:

3.4.2.7.1. Test or project identifier.

3.4.2.7.2. Special considerations (e.g., flight restrictions).

3.4.2.7.3. References to include review of previous similar test projects and lessons learned.

3.4.2.7.4. When applicable, other essential range safety criteria such as approved test areas, test items, danger areas, safety instrumentation requirements, safety footprint development methodology, etc.

3.4.2.8. **(Added-96TWSUP)** An example of a 96 TW/SEU published Safety Appendix is shown in **Attachment 2.** The format may vary depending on the complexity and nature of the test under consideration. The SRB Chairperson will determine the appropriate format.

3.4.2.9. **(Added-96TWSUP)** As a general rule, 96 TW SRBs are documented in the SRBS format shown in **Attachment 4**. Additional topics for consideration in the SRBS include brief background material, a Design Safety Certification statement (usually provided by 96 TW/SES), safety footprint development methodology and notes of all topics that were discussed. The SRBS may also include a discussion of items that were determined not to be a hazard as well as those identified as hazards. The rationale for determining the hazard minimizing procedures should be clearly documented in the SRBS (e.g., safety chase is required to inspect the store for security after each test point or terminate testing no later than one hour prior to sunset so as to preserve adequate ambient lighting for completing a visual battle damage check). Final approval of the SRBS resides with the Chief of Test and Range Safety or a designated representative.

3.4.2.9.1. **(Added-96TWSUP)** The SRBS will also contain the THA detailing the risk assessment for each hazard (e.g. assessment severity/probability components, causes, effects, mitigations, emergency procedures and any pertinent comments). The THA may be documented in the THA form (**AFTC Form 5000**) and included as an attachment to the SRBS.

3.4.3. Statement of Capability (SOC). The following wording must be included in any SOC that is transmitted to a customer when the safety review process is required:

3.4.3.1. "AFTC Safety Review: The proposed test/activity must be reviewed using the procedures contained in AFTCI 91-202, *AFTC Test Safety Review Policy* and any local supplements to this instruction. To support this review, safety planning must begin early in the program."

Chapter 4

RISK ASSESSMENT

4.1. General. Risk is defined as a combination of mishap severity and mishap probability. The overall risk level is the degree of risk assumed by leadership in allowing the proposed test to be accomplished in the manner described and under the conditions specified. Test teams will propose a risk assessment; independent reviewers will evaluate test unique hazards identified by the test team, assess proposed mitigations and corrective actions, and affirm or modify the test team's proposed overall risk level. Once the independent reviewers have agreed upon a risk level, they will document via the SRBS the overall risk level and recommendation to the TEA on whether or not to execute the test based on the SRB results. Test teams use system safety techniques, prior experience, legacy system research, lessons learned and overall engineering judgment to identify test hazards and assess risk by evaluating the credible outcome (mishap severity) of each hazard together with the associated probability of occurrence. The mishap severity and probability is then plotted on a Risk Assessment Matrix to determine the hazard's overall risk level. Although the goal is to minimize risk through good test and safety planning/review processes, the test may result in residual risk that must be directly accepted by the TEA in accordance with Section 6.1.

4.2. Determine Mishap Severity. Controls or minimizing procedures can be used to reduce the severity or probability of the hazard (**Chapter 3**). The mishap severity category is a qualitative assessment of the most reasonable credible mishap consequence that could occur with all mitigations in place. For activities at AFTC organizations, the mishap severity categories are shown in **Table 4.1** The assessment should incorporate engineering judgment and/or past experience with similar tests or systems and is often assessed with no mitigations and then reassessed with all minimizing procedures and corrective actions in place. The severity is assigned based on the system level consequence of total direct cost and severity of injury/occupational illness or equipment loss or damage. Descriptive definitions should be used as the primary criteria for assessing mishap severity. However, quantitative values may be used for higher cost SUT. Quantitative values for mishap severity listed in **Table 4.1** may be adjusted to match current guidance specified in AFI91-204, *Safety Investigations and Reports*.

Table 4.1. Mishap Severity Definitions.

Mishap Severity	Level	Descriptive	Quantitative	Mishap Class
Catastrophic	**1**	**Loss of life, aircraft, facility, or expensive system.**	**> $2M**	**A**
Critical	**2**	**Severe injury, lengthy hospital stay, or permanent injury. Severe aircraft, equipment or property damage.**	**$500K - $2M**	**B**
Marginal	**3**	**Minor injury, requiring medical lost work days, but no permanent injury. Minor damage.**	**$50K - $500K**	**C**
Negligible	**4**	**Superficial injury, little or no first aid required. Incidental, less than minor damage.**	**< $50K**	**D/E**

NOTES:

1. Use values listed in AFI91-204 for definitive guidance.

2. Environmental impact is assessed independent of the test risk and is documented on an AF Form 813 per AFI 32-7061 or 32 CFR Part 989.3 (d).

4.3. Determine Mishap Probability. After hazards have been identified and mitigation measures have been assessed and documented, the safety reviewers will subjectively assess the mishap probability. The mishap probability level should qualitatively and/or quantitatively measure the likelihood of the mishap occurring due to personnel error, environmental conditions, design inadequacies, procedural deficiencies, or system/subsystem component failure or malfunction. The assessment should incorporate engineering judgment and past experience with similar tests or systems with all minimizing procedures and corrective actions in place. If available, the test team and safety reviewers should consider the system safety analysis results from the contractor or system program office in order to understand areas of known concern. For operations where there is a well-developed database or sophisticated modeling/simulation, probabilities may be expressed quantitatively as 1 x 10-4, 3.8 x 10-6, etc. However, for developmental testing, the ability to compute numeric failure probability values with confidence is difficult because these activities involve new, complex, and often unproven systems. Therefore, **Table 4.2** also contains descriptive probability definitions (along with some example descriptive statements) that should be used as a standard to consistently assess mishap probability for all AFTC test activities.

4.3. **(96TWSUP) Table 4.2 Footnote.** Event is defined as the operation being assessed or a specific test point or configuration

Table 4.2. Mishap Probability Definitions.

Probability	Level	Descriptive	Quantitative (Probability of
Frequent	**A**	**Very likely to occur (e.g., test exceeds design limits or mishap occurred during similar testing, etc.)**	**$> 10^{-1}$**
Probable	**B**	**Likely to occur (e.g., test at design limits or mishap almost occurred during similar testing)**	**$< 10^{-1}$ but $> 10^{-2}$**
Occasional	**C**	**Some likelihood to occur, but not expected**	**$< 10^{-2}$ but $> 10^{-3}$**
Remote	**D**	**Unlikely to occur (e.g., test activity approaching design limits and done before with no problems encountered)**	**$< 10^{-3}$ but $> 10^{-6}$**
Improbable	**E**	**Rarely occurs (e.g. test activity within design limits and covered under normal operational procedures)**	**$< 10^{-6}$**
1 - Event may be defined in local supplements to this instruction.			

4.4. Risk Assessment Matrix. The risk assessment matrix, shown in **Figure 4.1**, is a tool for assessing mishap risk of test hazards as documented in safety planning documents. The risk categories are discretely divided into four shaded regions to distinguish between NEGLIGIBLE (green dotted), LOW (green), MEDIUM (yellow), and HIGH (high) risk levels. The correlation of approval authorities with the assigned overall risk level is discussed in **Chapter 6**. Despite the discrete distinction between each risk level, safety reviewers are reminded of the subjective nature of their assessment as each member incorporates engineering judgment and/or past experience with similar tests or systems into their risk level assessment. The use of the matrix defined in **Figure 4.1** and locally developed Test Safety Review Processes defined in supplements to this instruction are in accordance with AFI 91-202, *The US Air Force Mishap Prevention Program*, AFMC Sup, Chapter 13.

Figure 4.1. Risk Assessment Matrix.

Probability of Mishap Occurring During the Test	Mishap Severity Category			
	Catastrophic–I	**Critical–II**	**Marginal–III**	**Negligible–IV**
Frequent (A)	1	3	7	13
Probable (B)	2 **HIGH**	5	9	16
Occasional (C)	4	6 **MEDIUM**	11	18
Remote (D)	8	10	14 **LOW**	19
Improbable (E)	12	15	17	20 **NEGLIGIBLE**

4.5. NEGLIGIBLE Risk. The negligible risk assessment reflects a subset of LOW risk applicable to activities that either are or equivalent to normal or routine operations. The NEGLIGIBLE risk category is defined as hazards where the severity and probability assessments fall in the Negligible Severity column and Occasional, Remote, or Improbable Probability rows on the Risk Assessment Matrix. Due to the subjective nature of any risk assessment, an overall assessment greater than negligible for these blocks could still be appropriate.

4.5.1. For the severity category to be Negligible, the consequences of a mishap attributable to test activities must be less than minor injury or system damage. For personnel, the impact of the injury or illness equates to no work days lost. For equipment or facilities, less than minor damage equates to losses less than $50,000 (or current Class D definition). If the test team or reviewers identify test unique hazards that warrant a THA document, then an overall risk category of NEGLIGIBLE is not appropriate.

4.5.1.1. **(Added-96TWSUP)** A negligible risk assessment may be issued for test in which no test unique hazards are identified.

4.5.2. Examples include: ride-along data collection points, special instrumentation checkouts, form-fit-function checkouts of non-critical hardware/software, sensor or system tests, or logistics testing activities that do not directly affect the airworthiness of an aircraft or performance of a test facility nor are they required for hazard avoidance.

4.6. Determine Overall Risk Assessment. An overall risk level assessment is accomplished after all hazards to the test have been identified and mitigations are clearly defined and documented in accordance with Section 3.4. Hazards that are unique to the test will be documented on a *Test Hazard Analysis* Form (AFTC Form 5000), or in a locally developed format that captures the information required in AFI 91-202, AFMC supplement. Hazards associated with normal operation and maintenance may be documented in a locally produced BHA form. Plot the combination of mishap severity and probability on the Risk Assessment Matrix for each hazard. Once all the individual hazards are plotted, the test team will discuss the safety aspects of the plan and propose an overall project risk level. Project risk will be no lower than the highest assessed risk from the THAs. A detailed explanation of THAs and BHAs is discussed in Section 3.4, Test Package Documentation.

4.6.1. THA Risk Assessment. The test team may assess the pre- and post-mitigation mishap severity category and probability level by plotting both on the Risk Assessment Matrix (**Figure 4.1**). This provides a comparison between initial and residual risk levels to evaluate the adequacy of safety measures and best available solution. Test teams and safety reviewers should note that although risk mitigation in the safety plan may not change the assessed severity and probability, it will still reduce the actual risk. The residual risk level determined by the test team for each THA acts as a proposal for the independent safety reviewers to affirm or adjust as necessary.

4.6.2. Overall Risk Assessment. The test team will propose an overall risk level for the test as determined by procedures discussed in this section. During the safety review phase (outlined in **Chapter 5**), the independent safety reviewers will have a general discussion of the test, identified hazards, and associated mitigations to generate opinions on the residual risk. The discussions should be candid and result in a general agreement by the SRB, although disagreements may occur. Safety reviewers will weigh the control measures in place (mitigation steps), their experience with the types of tests, and the SUT to assess the overall residual risk. The cumulative risk may (and frequently does) exceed the assessed risks for all THAs individually. However, the overall risk cannot be lower than the risk associated with any individual THA. The safety reviewers must also consider the complexity of the test, the potential for safety-related “unknown unknowns”, and their own experience with similar test activities. By using the Risk Assessment Matrix (**Figure 4.1**) and referencing the overall risk level descriptions, shown in **Table 4.3**, each safety reviewer should assess overall risk and provide justification for their assessment. The overall risk assessment must be documented in the safety plan.

Table 4.3. Overall Risk Level Assessments.

Assessment	Description and Implication
HIGH Risk	Tests or activities that present a significant risk to personnel, equipment, and/or property after all precautionary measures have been taken.
MEDIUM Risk	Tests or activities that present a greater risk to personnel, equipment, and/or property than normal after all precautionary measures have been taken.
LOW Risk	Test or activities that present a little/no greater risk than normal operations (such as operating the system using approved procedures) after all precautionary measures have been taken. Routine supervision is appropriate.
NEGLIGIBLE Risk	Activities that either are or equivalent to normal or routine operations. representative.

4.6.2.1. In some situations, sufficient information may not be available to complete a risk assessment. The Test Safety Office of each AFTC organization will determine a course of action to develop resolution and may reconvene the safety reviewers to perform the assessment at a later date.

4.6.2.2. If appropriate, the risk may be assessed separately for AFTC and non-AFTC assets, for different phases of the test projects, or for individual test events. The overall risk for the test project is still based on the highest level of risk assessed on any of the tests, but the project can have split risk assessments. For example, an overall HIGH risk may be assigned for a test project which includes flight envelope expansion, but a subset of that testing may be assessed as MEDIUM or even LOW. If this is the case, the test points in each risk category will be clearly identified in the safety planning.

4.7. Elevated Risk Activities. Certain tests conducted at AFTC organizations have demonstrated a higher than normal risk due to the inherent hazards involved. However, if the analysis of test activities clearly indicates that the predicted performance (flying qualities, pilot induced oscillation susceptibility, flutter margin, loads margin, etc.) is well within acceptable levels, the test point need not be considered elevated risk. This may be especially true if the analysis model has been validated through other simulation or test activity.

Chapter 5

TEST SAFETY REVIEW PHASE

5.1. Safety Review Preparation. In preparation for an independent safety review, test teams should perform the following:

5.1.1. Determine the type of safety review (examples in **Paragraph 5.2.2**) and consult Wing/Complex Test Safety office for concurrence.

5.1.2. Evaluate the probability and severity category for each THA or BHA (**Chapter 4**). Provide to the safety reviewers the proposed overall risk level and any test points or test phases which may have a lower risk than the overall risk level (if they exist). Include the rationale for the varying risk levels. The proposed risk level(s) will be considered during the independent safety review.

5.1.3. Develop a list of safety reviewers following guidance in Section 2.3

5.2. Safety Review. The purpose of the Safety Review phase is to allow an independent team to formally review the test unit's safety planning to ensure that all test hazards have been identified and sufficiently mitigated, affirm or modify the residual risk, determine the overall risk level of the test and recommend to the TEA whether or not to execute the test. The documentation from this phase should reflect a suitable level of clarity and maturity for the TEA to make an informed decision on whether to proceed with test execution. The Wing/Complex Test Safety office is the focal point for the Safety Review phase.

5.2.1. Objectives:

5.2.1.1. Ensure appropriate test hazards associated with the test activity are identified.

5.2.1.2. Ensure the proposed risk control measures sufficiently mitigate (minimize or eliminate) the hazards caused by the test/activity to an acceptable level.

5.2.1.3. Assess and recommend an appropriate residual risk level for the test/activity.

5.2.1.4. Ensure the safety plan clearly and adequately provides enough information to support an approval decision by the TEA.

5.2.1.5. **(Added-96TWSUP)** Ensure that any test conducted through the 96 TW will be accomplished with all due respect to the safety of the aircrew, activity participants, facilities, resources involved and the general public.

5.2.1.6. **(Added-96TWSUP)** Provide a forum by which 96 TW customers, 96 TW/SE, 96 OG/CC, 96 TW/ CC and AFTC/CC are assured the test will be conducted as safely as possible.

5.2.2. Types of Independent Safety Reviews. Below are four types of independent safety reviews that may be used to complete the safety review phase. The Wing/Complex Test Safety office may advocate additional types of reviews as defined in local supplements to this instruction. The test team will review relevant documentation and propose a review type to the Test Safety office, who will make the final determination. The four types of independent safety reviews are:

5.2.2.1. Formal Safety Review Board (SRB)**.** This is a meeting attended by independent safety reviewers and project personnel, and is chaired by a designated Wing/Complex Test Safety office representative. The decision to conduct a Formal SRB vs an Electronic Safety Review (ESR) or Combined Technical Review Board (TRB)/SRB is based primarily on the test plan size, complexity, maturity of test item/methodology, and expected risk level and is determined by the Test Safety Office. To the maximum extent possible, independent safety reviewers chosen for the SRB should be the same individuals that served as independent reviewers for the technical review. This is to ensure continuity of information regarding test methodology is preserved throughout the review and approval process and should result in a more insightful and thorough SRB.

5.2.2.2. Combined TRB/SRB**.** For those tests that are easily understood, less complex, or lower in risk, the test team may request a combined TRB/SRB in lieu of separate technical and safety reviews to minimize impact to resources and shorten the timeline. Teams should contact the Test Safety Office for final determination on this course of action. Teams will ensure that the test plan is sufficiently mature for safety review prior to the combined TRB/SRB.

5.2.2.3. Electronic Safety Review (ESR)**.** The ESR is a formal safety review of test packages by the SRB that occurs without a meeting. The test package is typically distributed electronically and reviewed in parallel by the safety reviewers. An ESR is appropriate when test activities are readily understood by reviewers, tend to be less complex, and are lower in risk.

5.2.2.3.1. **(Added-96TWSUP)** An electronic streamlined risk assessment process (96 TW Livelink workflow) exists for evaluating tests that follow Standard Operating Procedures (SOP) in conjunction with the TD/MOT/CONOPS for facilities such as J-PRIMES and GWEF. This process may be used at the discretion of the SRB Chairperson after review of the MOT/ CONOPS for the specific activity. Additional electronic streamline risk assessment processes will be approved by 96 TW/SEU as appropriate to support facility or laboratory type tests. These streamline processes may also be issued to provide the risk assessment to the TEA and provide a means for approval and capturing the TEA signature signifying test approval.

5.2.2.4. Negligible Risk Review (NRR)**.** An NRR is a streamlined technical and safety review process applicable to a subset of low risk tests as indicated on **Figure 4.1**, Risk Assessment Matrix. Test activities that either are or equivalent to normal or routine operations (e.g., incidental to another routine flight activity or test) are excellent candidates for an NRR process since the risk is effectively the same as the operational risk. During risk assessment if all identified hazards fall within the negligible risk area in **Figure 4.1**, a NRR can be accomplished. A minimum of two test safety personnel must review the proposed assessment and at least one of them must have experience in the area being assessed as determined by the Chief of Test Safety.

5.2.2.4.1. NRR Qualification. NRR qualification of a test project should be proposed by the test team to either the TSO (if the TSO is outside the Wing/Complex Test Safety Office) or Wing/Complex test safety officer to make a preliminary risk assessment before forwarding to the Wing/Complex test safety office. The Test Safety Office will make the final determination based on the following criteria:

5.2.2.4.1.1. The risk level for the test activity must be assessed as negligible and fall within the hashed blocks in the Risk Assessment Matrix (see **Figure 4.1**). Examples of these activities are listed in **Paragraph 4.5.2**

5.2.2.4.1.2. Testing will adhere to normal operating procedures and existing risk control measures as defined in the approved flight manual(s), technical orders, test facility procedures, and/or operational guidance/instructions (e.g., Air Force Instructions, Air Force Materiel Command Instructions, and Air Force Test Center Instructions).

5.2.2.4.1.3. GMPs are allowed only to the extent that they clarify or further restrict already existing guidance. If the test team or reviewers identify test unique hazards that warrant a THA document, then the NRR process is not appropriate.

5.2.2.4.1.4. Routine and existing aircrew/operator training, qualification, and proficiency are sufficient to perform the test activity, test or maneuver.

5.2.2.4.1.5. Test procedures do not involve the use of abnormal or emergency procedures, checklists or configurations.

5.2.2.4.1.6. For flight test, the SUT has no airworthiness impact, such that a failure or malfunction of the SUT would cause the use of abnormal or emergency procedures to safely recover the aircraft.

5.2.2.4.2. Each Wing/Complex may define a NEGLIGIBLE risk review and approval process in a local supplement to this instruction. If defined locally, the NRR process will comply with NRR qualification guidance in this Chapter and the approval coordination path defined in **Table 6.1**

5.2.2.4.2.1. **(Added-96TWSUP)** A 96 TW Negligible Risk Review will follow the same general process of any Test Safety Review. Test teams are required to generate a 96 TW TD/MOT/CONOPS with associated safety plan as appropriate. Test documentation will be delivered to 96 TW/SEU for review. The SRB Chairperson will determine the appropriate review and ensure the integrity of the Test Safety Review Process IAW AFTCI 91-202 and publish a response in the form of a Safety Appendix, MFR or other appropriate means to document the safety review indicating the risk assessment.

5.3. (Added-96TWSUP) SRB Schedule. The SRB will be scheduled after the test or training organization has completed their technical review of the MOT or CONOPS. The SRB should be held as late in the planning cycle as possible. Any changes to the MOT or CONOPS after the SRB has been completed could result in the requirement to re-accomplish the Test Safety Review Process resulting in the re-evaluation of the previously defined risk assessment and/or re-convening of the SRB and/or additional review by the Test Execution Authority (TEA). A typical SRB timeline is shown in **Table 5.1** This schedule will vary depending on test complexity, test safety issues, etc. Streamlined test planning guidance is contained in the 96 TW Plan 70, *Crisis/ Command Directed Rapid Response Testing*. Re-convening of the SRB or review by the TEA will be done at the discretion of the SRB chairperson or TEA, respectively.

Table 5.1. (Added-96TWSUP) SRB Timeline.

Timeline	Events
SRB-5-10 workdays	The SRB will normally be scheduled between 5 and 10 workdays of receipt of all necessary documentation. Documentation* will include TD with MOT/CONOPS, flight profiles, Recommended Flight Clearance (RFC), mission summaries, system descriptions, munitions types/ quantities, test item recovery or dud analysis requirements, test areas, targets, AFTC Form 5000 (THA) and other information as required.
SRB-2 workdays	The SRB chairperson will contact the SRB requester with any concerns not addressed.
SRB	Key members as approved by the SRB Chairperson and defined in **Paragraph 5.4** present and prepared to discuss the hazards associated with the documented activity.
SRB + 8 workdays	Safety Appendix signed and ready for publication.
	* – These documents and information are not required for all SRBs. Draft versions are satisfactory. The SRB chairperson will determine if the required documentation and data is sufficient to proceed with SRB scheduling.

5.4. **(Added-96TWSUP) SRB Membership.** Membership will be determined by the type of test being conducted, the responsible test organization and where the test is being conducted. SRB membership for each of these types of tests will be **IAW AFTCI 91-202, Paragraph 2.3; SRB members may fulfill multiple roles as defined and approved by the SRB chairperson.**

5.4.1. **(Added-96TWSUP)** The required SRB core participants for all flight tests include:

5.4.1.1. **(Added-96TWSUP)** Test and Range Safety Engineer

5.4.1.2. **(Added-96TWSUP)** Flight Safety Representative

5.4.1.3. **(Added-96TWSUP)** Test Engineer, Test Manager or equivalent

5.4.1.4. **(Added-96TWSUP)** Test Aircrew Representative

5.4.1.5. **(Added-96TWSUP)** All tests using any ammunition, flares, explosive devices, aircraft store/weapons system or directed energy weapon will also require a Weapons Safety Representative and Explosive Ordnance Disposal (EOD) representation as necessary. Any Air Force compatibility test conducting carriage or release of a new store, suspension hardware or aircraft/store configuration; expanding the carriage or release envelope of an existing store; or expanding the carriage or release envelope of an existing aircraft/store configuration will require a SEEK EAGLE Program Representative as applicable or from an equivalent organization. Deviations will be coordinated between the required core member and the SRB chairperson prior to the SRB. The SRB chairperson will make the final determination to continue the scheduled SRB or reschedule considering the coordination discussion and test complexity.

5.4.2. **(Added-96TWSUP)** The required SRB core participants for all ground tests include:

5.4.2.1. **(Added-96TWSUP)** Test and Range Safety Engineer

5.4.2.2. **(Added-96TWSUP)** Test Engineer, Test Manager or equivalent

5.4.2.2.1. **(Added-96TWSUP)** All tests using any ammunition, flares, explosive devices, aircraft store/weapons system or directed energy weapon will also require a Weapons Safety Representative and EOD representation as necessary. Deviations will be coordinated between the required core member and the SRB chairperson prior to the SRB. The SRB chairperson will make the final determination to continue the scheduled SRB or reschedule considering the coordination discussion and test complexity.

5.4.3. **(Added-96TWSUP)** Additional required participants will be as necessary depending on the test procedures, test items, resources, and/or other test specifics or peculiarities.

5.5. **(Added-96TWSUP) SRB Member Responsibilities.** All SRB members must attend the SRB having reviewed the MOT/CONOPS, have an understanding of the identified hazards and any proposed minimizing procedures, and submitted any THA forms as necessary. If the membership has not been adequately prepared, the SRB chairperson may cancel the SRB and reschedule at such time the membership is prepared to conduct the SRB.

5.5.1. **(Added-96TWSUP)** 96 TW Test and Range Safety Officer (96 TW/SEU)

5.5.1.1. **(Added-96TWSUP)** Establish contact with the test requester, test engineer and/or test pilot/ aircrew member in order to initiate a preliminary review of the test.

5.5.1.2. **(Added-96TWSUP)** Prior to the SRB, review the MOT or CONOPS to ensure all topics pertinent to the test are addressed.

5.5.1.3. **(Added-96TWSUP)** Review appropriate test documentation and THA archives to determine if there are any applicable topics to consider at the SRB.

5.5.1.4. **(Added-96TWSUP)** Complete, as needed, AFTC Form 5000 (THA) for each newly identified test-unique hazard.

5.5.1.5. **(Added-96TWSUP)** Schedule the SRB IAW the scheduling timeline detailed in **Table 5.1** All efforts shall be used to notify participants between five and ten workdays prior to the SRB. This will allow all members to arrange their schedules and review applicable documentation.

5.5.1.6. **(Added-96TWSUP)** Document the safety requirements in a format that clearly conveys the identified test hazards, minimizing procedures and risk assessment. All documented test unique hazards and mitigating/minimizing procedures should be included in the Test Approval Brief (TAB).

5.5.1.7. **(Added-96TWSUP)** Document the minutes of the SRB in a format that details how the risk assessment for each identified hazard was determined; this format includes detailing the hazard causes, effects and the assessment components. The SRBS will document all subjects discussed at the SRB. The SRBS will detail not only the aspects of the test that were identified as being hazardous, but also those that were rejected and the reasons for that determination.

5.5.1.8. **(Added-96TWSUP)** Document any action items completed or additional discussion points that have occurred after the SRB for inclusion in the Safety Appendix. Any action items or additional discussion point may delay the publication of the Safety Appendix beyond the typical timelines identified in **Table 5.1**

5.5.1.9. **(Added-96TWSUP)** Ensure identified ISRs and TEs have coordinated and concurred on the final safety plan as documented in the Safety Appendix. If the ISRs have non-concurred, justification and reasoning behind non-concurrence will be documented in the SRBS and briefed prior to TEA approval decision.

5.5.2. **(Added-96TWSUP)** 96 TW Systems Safety Engineer (96 TW/SES)

5.5.2.1. **(Added-96TWSUP)** For USAF munitions and test items, 96 TW/SES responsibilities include:

5.5.2.1.1. **(Added-96TWSUP)** Review the system design or modification to identify system design hazards.

5.5.2.1.2. **(Added-96TWSUP)** Provide a Design Safety Certification statement to be incorporated into the SRBS.

5.5.2.1.3. **(Added-96TWSUP)** As applicable, participate in the early system design phases to identify and eliminate design hazards.

5.5.2.1.4. **(Added-96TWSUP)** Provide the SRB with an assessment of potential hazards associated with the weapon system or munition.

5.5.2.1.5. **(Added-96TWSUP)** Identify any potential hazards associated with weapons design regarding live weapon safe separation from the parent aircraft.

5.5.2.1.6. **(Added-96TWSUP)** Complete and/or provide inputs to AFTC Form 5000 (THA) for each identified system design hazard as required.

5.5.2.2. **(Added-96TWSUP)** For non-USAF munitions and test items, provide systems safety expertise as needed and dependent upon the data and time available for analysis. Test item contractor, test range customer and/or other service systems safety organization may provide the Design Safety Certification statement and other required inputs in lieu of 96 TW/SES.

5.5.3. **(Added-96TWSUP)** 96 TW Flight Safety Representative (96 TW/SEF)

5.5.3.1. **(Added-96TWSUP)** Review the flying objectives and procedures as defined in the MOT/CONOPS prior to the SRB and provide safety of flight expertise relative to those objectives/procedures.

5.5.3.2. **(Added-96TWSUP)** Ensure mishap accountability is addressed at the SRB and documented in the SRBS.

5.5.3.3. **(Added-96TWSUP)** Complete, as needed, AFTC Form 5000 (THA) for each newly identified safety of flight related test-unique hazard.

5.5.4. **(Added-96TWSUP)** 96 TW Weapons Safety Representative (96 TW/SEW)

5.5.4.1. **(Added-96TWSUP)** Review the weapons objectives and procedures as defined in the MOT/ CONOPS prior to the SRB and provide weapons safety expertise relative to those objectives/procedures.

5.5.4.2. **(Added-96TWSUP)** Ensure operations involving ammunition, flares, explosive devices, aircraft stores/weapons systems or directed energy weapons are conducted IAW applicable DoD and AF requirements.

5.5.4.3. **(Added-96TWSUP)** Verify explosive safety training qualifications on project personnel required to either handle munitions or perform explosive operations.

5.5.4.4. **(Added-96TWSUP)** Review, coordinate and approve all munitions tie-down procedures (trailer transportation).

5.5.4.5. **(Added-96TWSUP)** Identify any hazards associated with the recovery and analysis (x-ray, sawing and disassembly) of dud or misfired munitions.

5.5.4.6. **(Added-96TWSUP)** Review, coordinate and approve all munitions assembly/disassembly procedures, modification of explosive items, EOD Safe Handling and Recovery Procedures (SHARP), technical data packages and hung ordnance procedures.

5.5.4.7. **(Added-96TWSUP)** Verify and approve correct storage requirements if non-standard (i.e., items not stored in munitions storage area or requiring licensed locations on range.)

5.5.4.8. **(Added-96TWSUP)** Complete, as needed, AFTC Form 5000 (THA) for each newly identified weapons or explosive test-unique hazard.

5.5.5. **(Added-96TWSUP)** 96 TW Occupational Safety Representative (96 TW/SEG)

5.5.5.1. **(Added-96TWSUP)** Review the objectives and procedures as defined in the MOT/CONOPS prior to the SRB and provide occupational safety expertise relative to those objectives, procedures and locations.

5.5.5.2. **(Added-96TWSUP)** Provide inputs to AFTC Form 5000 (THA) for each identified occupational safety related hazard, as required.

5.5.6. **(Added-96TWSUP)** Test Engineer, Test Manager or equivalent

5.5.6.1. **(Added-96TWSUP)** Provide Test and Range Safety with all details of the test as outlined in **Table 5.1** of this supplement along with an associated safety plan.

5.5.6.2. **(Added-96TWSUP)** Identify any test unique hazards associated with all operations involved with the test. This will include but not be limited to:

5.5.6.2.1. **(Added-96TWSUP)** Weapons build-up, weapon transportation, weapon handling, and weapon loading.

5.5.6.2.2. **(Added-96TWSUP)** Target locations, target movement (e.g., blocks), target construction, target disposition/clean up, etc.

5.5.6.2.3. **(Added-96TWSUP)** Instrumentation requirements that may require personnel inside the weapons safety footprint or other defined hazard area.

5.5.6.3. **(Added-96TWSUP)** Complete, as required, AFTC Form 5000 (THA) for each newly identified test-unique hazard proposing the hazard minimizing procedures.

5.5.7. **(Added-96TWSUP)** Test Pilot or Aircrew Representative. Test aircrews need to be involved with test programs from program inception. Aircrew inputs need to be made throughout the MOT development process. In conjunction with the test engineer, the test aircrew is required as needed to perform THA by identifying test-unique hazards, determining causes and effects and developing minimizing procedures. AFTC Form 5000 (THA) should be completed by the test engineer and test aircrew and delivered to Test and Range Safety prior to the SRB per **Table 5.1** The test aircrew and/or test engineer will brief these THAs at the SRB.

5.5.7.1. **(Added-96TWSUP)** Review the MOT/CONOPS to ensure all aircraft performance requirements are within the capabilities of the aircraft.

5.5.7.2. **(Added-96TWSUP)** Ensure the test is conducted IAW current USAF, AFTC, 96 TW and/or other applicable instructions.

5.5.7.3. **(Added-96TWSUP)** For any test points near, at or exceeding the aircraft envelope, identify any hazards and develop minimizing procedures to be presented at the SRB.

5.5.7.4. **(Added-96TWSUP)** Advise the SRB on any flight requirements that exceed the aircraft flight design limits or may put the aircraft/aircrew in a hazardous situation.

5.5.7.5. **(Added-96TWSUP)** For store release missions, ensure all pre-launch switch settings can be completed within the test area. If not, identify to the SRB any hazards that may be associated with switch settings.

5.5.7.6. **(Added-96TWSUP)** Discuss hung ordnance procedures, if applicable. Where local standard hung ordnance procedures are not appropriate, develop, coordinate and brief alternative procedures for the specific program.

5.5.7.7. **(Added-96TWSUP)** Identify any condition unique to the proposed test profiles (for example: weather, bird strike, low altitude flight) that may put the aircraft/aircrew in an elevated risk situation. Present the appropriate minimizing procedures to the SRB.

5.5.7.8. **(Added-96TWSUP)** When possible, coordinate with the SEEK EAGLE Program representative concerning all hazards associated with maneuvering with the aircraft/store configuration prior to the SRB.

5.5.8. **(Added-96TWSUP)** The SEEK EAGLE Program (AFI 63-104) Representative. The following requirements apply to SEEK EAGLE programs.

5.5.8.1. **(Added-96TWSUP)** Provide the SRB with documentation generated in support of the planned testing, recommended flight clearance, risk assessment, engineering letter or certification recommendation.

5.5.8.2. **(Added-96TWSUP)** Provide the SRB with mission summaries associated with the planned test.

5.5.8.3. **(Added-96TWSUP)** Provide a store separation assessment for each ejection, release, dispense or launch of a store or system. The store separation assessment should also be accompanied by relevant aircraft loads assessment (e.g., dynamic response, kick-back forces), as necessary.

5.5.8.4. **(Added-96TWSUP)** Provide safe escape/safe separation analysis for live weapon releases/ launches. Identify any hazards associated with safe escape/safe separation for the parent or chase aircraft as requested by 96 TW/SEU.

5.5.8.5. **(Added-96TWSUP)** Identify any potential hazards associated with maneuvering with the test aircraft/store configuration (aircraft loads, store loads, aircraft flutter, aircraft handling). All efforts should be made to coordinate hazards associated with maneuvers with aircrews prior to the SRB.

5.5.8.6. **(Added-96TWSUP)** Identify any potential hazards associated with operation of all test aircraft/ store configurations which may adversely impact safety through unintended electromagnetic compatibility or electromagnetic interference (EMC/EMI) issues.

5.5.8.7. **(Added-96TWSUP)** Complete and/or provide inputs to AFTC Form 5000 (THA) for each flight clearance identified hazard in the area of their expertise, as required.

5.5.9. **(Added-96TWSUP)** EOD Representative (96 CES/CED)

5.5.9.1. **(Added-96TWSUP)** Provide the SRB with explosive operations expertise.

5.5.9.2. **(Added-96TWSUP)** Identify any hazards associated with the disposal or recovery of unspent, dud or misfired munitions.

5.5.9.3. **(Added-96TWSUP)** Identify any hazards associated with the recovery and analysis (x-ray, sawing) of unspent, dud or misfired munitions.

5.5.9.4. **(Added-96TWSUP)** In conjunction with weapon designer inputs, identify wait times for dud recovery.

5.5.9.5. **(Added-96TWSUP)** Identify any unique hazards which are involved with safing, downloading and transportation of weapons which are recovered during an aircraft emergency hung store situation.

5.5.9.6. **(Added-96TWSUP)** Complete and/or provide inputs to AFTC Form 5000 (THA) for each identified explosive hazard, as required.

5.5.10. **(Added-96TWSUP)** Bioenvironmental Engineer (96 AMDS/SGPB).

5.5.10.1. **(Added-96TWSUP)** Provide the SRB with bioenvironmental expertise.

5.5.10.2. **(Added-96TWSUP)** Identify any health hazards associated with radiation exposure including, but not limited to, ionizing radiation, radio frequency (RF) radiation, laser radiation and microwave radiation.

5.5.10.3. **(Added-96TWSUP)** Identify and advise the SRB on health hazards associated with chemical or other hazardous material exposures and the use of personal protective equipment (PPE).

5.5.10.4. **(Added-96TWSUP)** Identify and advise the SRB on heat stress issues.

5.5.10.5. **(Added-96TWSUP)** Identify and advise the SRB on potential health risks associated with tests affecting the environment.

5.5.10.6. **(Added-96TWSUP)** Identify and advise the SRB on potential audio health risks.

5.5.10.7. **(Added-96TWSUP)** Complete and/or provide inputs to AFTC Form 5000 (THA) for each identified bioenvironmental hazard, as required.

5.6. **(Added-96TWSUP) SRB Agenda.** The SRB is fundamentally a risk management effort. The SRB is neither a pre-TAB (dry run) nor a test planning meeting. The SRB can be in any form that effectively communicates test-related information to the members of the SRB. At the discretion of the SRB chairperson, the SRB may be convened and conducted using a variety of means to include meetings, email, telecon, VTC, etc. In a typical SRB, the test engineer will brief the planned systems under test; test objectives, test design, procedures, and any preliminarily identified test-unique hazards and proposed minimizing procedures. The SRB briefing should address, at a minimum, the following topics to a level of detail which supports a risk assessment.

5.6.1. **(Added-96TWSUP)** System(s) description.

5.6.2. **(Added-96TWSUP)** Test background to include test requester, purpose, scope, timeline, and resources involved.

5.6.3. **(Added-96TWSUP)** Test objectives.

5.6.4. **(Added-96TWSUP)** Method of test to include data collection, facilities, personnel, ranges, flight profiles and agencies involved.

5.6.5. **(Added-96TWSUP)** Any special test requirements.

5.6.6. **(Added-96TWSUP)** Missions summaries and highlights to include test design (if necessary), test approach precedence/justification, analogous systems/tests, supporting analysis, inherent risk mitigation and adopted risk mitigation.

5.6.7. **(Added-96TWSUP)** Test specific hazards and any potential minimizing procedures.

5.6.8. **(Added-96TWSUP)** All AFTC Forms 5000 (THA) for the test.

5.6.9. **(Added-96TWSUP)** Proposed risk assessment and conclusion.

5.7. (Added-96TWSUP) SRB Discussion. After the briefing, the SRB chairperson will lead a discussion to review whether hazards have been adequately identified and addressed. The SRB chairperson will also strive to discern any other potential test-unique hazards or safety problems. This discussion will emphasize the following areas of concern:

5.7.1. **(Added-96TWSUP)** General questions.

5.7.2. **(Added-96TWSUP)** Review existing AFTC Forms 5000 for test-unique hazards, adequate/ appropriate minimizing procedures, adequate emergency procedures and any comments.

5.7.3. **(Added-96TWSUP)** Complete or finalize any requirement for new THA forms as necessary.

5.7.4. **(Added-96TWSUP)** Consider any additional risk mitigation that could reasonably be applied.

5.7.5. **(Added-96TWSUP)** Consider lessons learned from previous testing that can be applied in such a way as to further minimize risk associated with current testing.

5.7.6. **(Added-96TWSUP)** Assign risk levels using the risk level assessment as outlined in AFTCI 91-202, Chapter 4.

5.7.7. **(Added-96TWSUP)** Summarize any open action items, risk levels associated with individual test parameters, exact wording of any restrictions placed on the test, reasons for restrictions and risk categories, and other items that will appear in the SA or SRBS.

Chapter 6

TEST SAFETY APPROVAL PHASE

6.1. Approval Authorities and Notification Levels. All activities conducted in accordance with **paragraph 1.6** require approval before beginning execution. The approval phase provides appropriate leadership the opportunity to make an informed risk acceptance and test approval decision based on the safety review and risk assessment completed in the safety review phase. The Test Execution Authority (TEA) for these activities is based on the overall risk level as outlined in **Table 6.1** Approval is defined as permission to conduct or participate in the test project or activity granted by the appropriate TEA. The TEA may require a Test Approval Brief (TAB) to assist in making an informed decision. Signature of the TEA on AFTC Form 5001 or equivalent, constitutes acceptance of the risk and approval to begin activities under the conditions set forth in the test package. A signed safety package does not authorize deviation from Air Force, AFMC, or AFTC instructions or directives.

6.1.1. **(Added-96TWSUP)** Signature of the delegated authority on the TD signifies test approval and risk acceptance. LiveLink is the official 96 TW archive for all test documentation and notification to the higher authority should include the link to all test documentation.

Table 6.1. Approval Process Coordination Path.

Organization Level	LOW Risk	MEDIUM Risk	HIGH Risk
Safety Office	Coord	Coord	Coord
Squadron CC (or equivalent)	Approve[1]	Coord	Coord
Group CC (or equivalent)	Info	Approve[1]	Coord
Wing/Complex CC	Not Required	Info	Coord
AFTC/SE	Not Required	Not Required	Coord
AFTC/CC	Not Required	Not Required	Approve[2]
HQ AFMC/SE and	Not Required	Not Required	Info

1. A deputy commander may assume the commander's role as the TEA.

2. High risk approval may be delegated in writing to the Test Wing/Complex Commander (CC). In the absence of the Test Wing/Complex CC, the vice commander can approve the testing; however, this cannot be further delegated. If delegated to a Test Wing/Complex CC, the AFTC/SE and AFTC/CC will be coordinated for 'Info' only.

6.2. LOW Risk Activities. The TEA for all LOW risk (including NEGLIGIBLE risk) test events is no lower than the responsible Squadron CC (or equivalent).

6.3. Elevated Risk Activities. Elevated risk activities are those that result in a residual risk level of MEDIUM or HIGH.

6.3.1. MEDIUM Risk Test Approval**.** Medium risk tests require approval of the Group CC or equivalent (minimum O-6 or civilian equivalent).

6.3.2. HIGH Risk Test Approval.

6.3.2.1. The AFTC/CC is the TEA for all HIGH risk test events. Final approval to execute HIGH risk test may be delegated in writing to the Test Wing/Complex CC.

6.3.2.2. If non-AFTC assets/personnel are involved, the asset owner must be notified of the high residual risk prior to test execution. Notification method will be established in local supplements.

6.3.2.2.1. **(Added-96TWSUP)** Non-AFTC assets/personnel owner will be notified in such a way that the risk is conveyed completely and accurately. Access to the TD and other applicable documentation will be provided as appropriate.

6.3.2.3. HQ AFMC/SE and AFMC/A3 must be notified of high risk tests prior to execution in accordance with AFI 91-202 AFMC Sup para 13.3.3.4. AFTC /SE will send this notification in conjunction with HIGH risk safety plan approval. The Wing/Complex safety office, or designee, will ensure AFTC/SE is notified within 24 hours when a test project has been approved for conduct as HIGH risk.

6.4. Test Approval Brief. The TEA or any other Commander on the Approval Coordination Path may require a TAB to assist in making an informed decision. A TAB should be an executive level meeting that provides a test project overview and highlights test unique hazards, mitigation measures, discussion points during the independent review (e.g., Formal SRB, ESR, Combined TRB/SRB), and any contention or disagreement by the independent board and the test team. The approval authority can attend the SRB and eliminate the potential need for a TAB. If a separate TAB is held and if slides are used, the slides will be archived with the test package documentation.

6.4.1. **(Added-96TWSUP)** The TAB is established to provide a management level review of test safety and provide final approval of the test; a TAB will not be scheduled prior to publication of the Safety Appendix. The TAB may be presented during a formal TAB review or during an informal or out-of-cycle meeting at the discretion of the TEA. This meeting is the forum where control decisions and the residual risk level identified at the SRB are accepted, rejected, minimized further or elevated to a higher authority for approval. The TEA approves a test only when all hazards have been reviewed, satisfactory planning and analysis have been conducted, and procedures have been established to allow accomplishment of the test with a high expectation of safe conduct. The TEA does not provide or approve airworthiness certification, flight clearances, design safety certification, or test safety criteria (for example, impact footprints or Flight Termination System certifications). These certifications must be obtained from appropriate sources and reviewed at the SRB. The TEA may authorize additional test

limitations or restrictions. Any limitations and/or restrictions which conflict with the requirements published in the Safety Appendix may result in the re-convening of SRB and publishing a new risk assessment based on the removal of those safety requirements. Any additional TEA stipulations, limitations, or restrictions will be captured in the TD for approval.

6.4.2. **(Added-96TWSUP)** The TEA will review/approve all tests, for which 96 TW is the ETO or PTO, tests that involve 96 TW possessed aircraft or flight by aircrews assigned to 96 TW, or when TEA review is otherwise requested. Questions regarding review criteria should be referred to the Test and Range Safety Office.

6.5. Acceptance of Safety Planning across AFTC

6.5.1. An AFTC test project which has been approved through an AFTC Wing/Complex's technical and safety review processes may be executed by a different, supporting, AFTC test wing/complex.

6.5.2. The originating test wing/complex will notify the supporting wing/complex when the technical and safety review processes are complete and the test project is approved for execution by the originating test wing/complex TEA. The originating test wing/complex will provide the supporting wing/complex with test and safety planning documentation required under the originating test wing/complex processes. The supporting wing/complex Test Safety office will review this documentation and may accept it as written, or may require additional safety review following their wing/complex supplement to this instruction. Differences will be resolved by equivalent TEAs from each wing/complex. The supporting wing/complex may then execute any assigned portion of a test project which has been approved to execute under the originating test wing/complex processes.

6.5.2.1. If changes are made to the safety package at the supporting wing/complex, such as additional GMPs due to range, complex or facility differences, the originating test wing/complex will be notified via a memorandum. The memo will include, as a minimum, the project title, additions to the package and rationale for the additions. If amendments are required for test or safety planning reasons, or an unexpected test event, the supporting wing/complex will provide the originating wing a copy of the amendment.

6.5.3. Test execution materials (e.g., test cards or mission decks) may be developed by either the originating or supporting test wing. The organization creating the mission materials will adhere to local guidance for formatting, content and approval. Mission materials will be approved by the executing organization in accordance with their local procedures.

Chapter 7

TEST EXECUTION PHASE

7.1. General. Risk management must be integrated and documented into all stages of T&E activities to identify test hazards, mitigating measures and acceptance/rejection of the residual risk by an appropriate TEA. The safety plan records due diligence in risk management and acceptance, and also communicates (e.g., provides a written copy of) hazards and mitigating measures to test personnel. The procedures, restrictions, and mitigations documented in the safety planning must be observed while conducting the test in order to maintain the accepted level of risk. Safety plan requirements take precedence over those specified in the test plan. The test package is a contract between the test team and the TEA.

7.2. Test Card/Test Period Directive Preparation and Approval

7.2.1. Test Cards/Test Period Directives/etc. are documents describing the test activity procedures in a step-by-step or checklist format. These documents are used by test teams to successfully complete test activities. They may be reused for multiple test projects but should not be overly general in documentation. Inherently, they should be a synopsis of operation, test and/or manufacturing technical data immediately available to reference for the test team in executing test activities effectively, efficiently and safely.

7.2.2. During test card or test period directive preparation, the test team will review applicable GMPs, THAs and BHAs to ensure the procedures comply with safety limits, procedural constraints or approved test plan requirements.

7.2.3. Test execution procedures, whether documented in test cards or another format, must be approved prior to use during testing. Test card approval levels will be documented in local Wing/Complex instructions.

7.2.3.1. **(Added-96TWSUP)** Test cards used for 96 TW activities will be approved according to the risk level assigned for the test points or test operations described. Test cards describing operations that fall under different risk assessments will follow the highest assessed risk test point included in those procedures. If the overall program or project is assessed at high or medium risk, but the test points to be executed are assessed at a lower risk, then test cards will be approved at the lower risk level. For example, if a program is assessed as medium risk, but the medium risk test points are not yet included in current test cards, then the test cards will be approved as low risk.

7.2.4. The order or sequence of the test cards may have a direct effect on the safety of a given test mission. Approved test cards, or “test decks”, may be reordered or re-sequenced without re-approval if there is no impact to the required buildup order or test safety. Test teams must ensure that test approach and build-ups, as defined or intended in the test and safety plans, are adhered to in all cases, and they should carefully analyze test point sequencing to avoid hidden pitfalls. Resequencing of test cards with a safety build-up as prescribed in the safety plan requires a safety review and amendment.

7.3. Test/Mission Execution Briefing. During the test/mission execution brief, the test team will address the procedures and restrictions specified in the safety plan. Test unique hazards applicable to the scheduled testing, risk minimizing procedures or controls, and go/no-go criteria must be briefed at the test/mission execution briefing. These can be captured in GMPs or THAs.

7.4. Unexpected Test Event

7.4. **(96TWSUP) 96 TW Weapons Safety will:** Be notified of all unexpected test events involving ammunition, flares, explosive devices, aircraft stores/weapons systems or directed energy weapons.

7.4.1. Unexpected test events that affect the continued safe execution of the test include, but are not limited to:

7.4.1.1. Unexpected or unplanned damage to the SUT or support equipment.

7.4.1.2. Exceeding safety of test limits.

7.4.1.3. Unfavorable departure from predicted simulation/analysis.

7.4.1.4. Occurrence of a hazard with an Improbable, Remote or Occasional probability of occurrence.

7.4.1.5. Failure of planned mitigations that allowed a hazard to occur.

7.4.2. If an unexpected test event occurs (actual or suspected), the test team will put the test on hold and consult with the Wing/Complex test safety office for confirmation of an unexpected test event. If confirmed, the Wing/Complex test safety office will provide notification of the event through the appropriate chain of command. Test points associated with the unexpected test event will be placed on hold, but if the test team and the safety office concur, other unrelated test points can continue. The SRB chair may also elect to reconvene the SRB to review and revalidate that all associated risks have been mitigated/addressed before testing can continue.

7.4.3. Once a recovery plan of action is determined, unexpected test events will be documented with a safety plan amendment (a memorandum format could be used). Testing of the suspended test points may be resumed upon approval of the appropriate change documentation as described in **Chapter 8**. An Unexpected Test Event amendment does not constitute a mishap investigation, if one is required by AFI 91-204.

7.5. Hazard Occurrence but Not an Unexpected Test Event. If a hazard occurs that is not considered an unexpected test event, it will be reported to the Wing/Complex test safety office as soon as practical (e.g., after post-test debrief). A memorandum should be added to the safety plan to document the hazard occurrence. This information could be of use to future test teams in their safety planning.

7.6. (Added-96TWSUP) Test execution will: Be IAW applicable range operating instructions to include but not limited to DoDI 3200.18 and AFMAN 13-212 and any locally developed supplements.

Chapter 8

CHANGES AND TIME LIMITS

8.1. Changes. It is not unusual for project changes to arise after receiving test approval. Unexpected results, overly restrictive controls in THAs or GMPs, hazards not previously identified or adequately controlled, and proposed changes in risk level all constitute reasonable grounds for changing safety planning. All project changes will re-accomplish the following test safety review process phases: safety planning, safety review, and approval. However, the scope of each phase may differ significantly from that of an original safety plan, depending on the changes and documentation method used.

8.2. Major Changes. Any potential change in risk level (higher or lower), major test plan change, changes to safety planning, and unexpected test events are considered major changes that affect test conduct or safety planning. Major changes require additional safety planning, independent safety review, and approval before continued testing with these changes incorporated.

8.2.1. Risk Level Change. During the course of testing, information may be obtained that potentially warrants a change in risk level. This could be an increase in the risk based on unexpected results or a decrease in risk level due to increased system maturity.

8.2.1.1. The approval authority for an increase in risk level will be based on the "new" risk level IAW **Chapter 6** (i.e. an upward change to HIGH risk requires AFTC/CC approval if not already delegated).

8.2.1.2. The approval authority for a decrease in risk level will be based on the "original" risk level IAW **Chapter 6** (i.e. a downward change from HIGH risk requires AFTC/CC approval if not already delegated).

8.2.1.3. Changes to testing approved with split risk levels. For amendments to test packages with split risk levels (see **paragraph 4.6.2.2**), the approval authority for the amendment will be based on the portion of the test package that is being changed. Information copies will be sent to the original approver.

8.2.1.3.1. For example, a test package has been approved as HIGH risk for test points over 800 KCAS and MEDIUM risk for all other test points. An amendment is submitted that only affects test points below 800 KCAS. The approval authority for the amendment corresponds to the MEDIUM risk TEA as IAW **Table 6.1**

8.2.2. Major Test Plan Change. The definition of major test plan change will be outlined in local supplements. Substantive changes to test objectives, technical approach, or test procedures may require an amendment to the safety plan. Individuals performing the final safety review should be the same as those from the original package, if available. For multi-discipline test plans, only the discipline(s) affected by the amendment need to be included for review along with an operations representative. If there is no change to the safety plan, then a memorandum format for the amendment could be used. The memo will be coordinated and approved per **Table 6.1** If there is a change to the safety plan, then follow **paragraph 8.2.3**

8.2.2. **(96TWSUP)** Changes are considered major if the change is outside/expands the scope of the MOT/CONOPS, if the assigned risk level has changed or is expected to change, if any published safety requirement is expected to be amended, or if the change introduces any additional test unique hazards not assessed by the SRB or increases the probability/severity of any previously identified hazard. The amended MOT/CONOPS should be forwarded to 96 TW/SEU for evaluation and recommendation for the type of SRB to be convened IAW AFTCI 91-202, Chapter 5.

8.2.3. Change to Safety Planning. Any change to the content of the safety plan is considered a change to safety planning. The desired changes could be more restrictive or less restrictive than the approved safety planning. Changes to safety planning will be accomplished via an amendment to the original safety package. Amendments will be approved using an AFTC Form 5001, or locally developed electronic or hard copy document which captures the appropriate coordination and approval signatures. For minor safety plan changes (e.g., the change is within the scope of the previously approved safety plan), a memorandum format for the amendment could be used. The memo will be reviewed by the appropriate safety reviewers, coordinated and approved per **Table 6.1**

8.2.4. Unexpected Test Event. Safety plan documentation following an unexpected test event will describe the occurrence of the event, summarize the cause(s) as they are understood by either analysis or hypothesis, and identify the test team's intended path for the resumption of testing. Once suspended for safety, only the TEA (or higher) can authorize resumption of testing.

8.3. Minor and Administrative Changes. Some changes to the approved test package may be classified as minor or administrative only and will be defined in local supplements. Minor test plan changes may include changing the flight conditions of test points, adding test points (provided the new conditions are within the approved envelope of test points), or deleting test points that are not a part of safety build-up. An administrative change to the test package clarifies information contained in the package and does not affect test conduct or safety planning. Locally approved procedures for documenting and approving minor or administrative changes may be defined in supplements to this instruction. The squadron commander (or equivalent) may be the approval authority for any changes not defined as Major Changes in Section 8.2.

8.3. (96TWSUP) Changes are considered minor if: The change does not alter the test procedures as defined in the MOT/CONOPS, affect the risk assessment, or if no new test unique hazards have been introduced. The amended MOT/CONOPS should be forwarded to 96 TW/SEU for evaluation and recommendation for the type of SRB to be convened if necessary.

8.4. Time Limit. As part of the RM process, safety plans will be reviewed at least every three years to ensure identified hazards and mitigation measures are appropriate and to incorporate any lessons learned. Baseline Safety Reports and USAF Test Pilot School standard curriculum event safety plans will be reviewed at least every four years. During this review, test teams will identify any new risks and mitigation measures; highlight key issues experienced since approval or the last review; and purge non-applicable guidance from the plan. Teams will document reviews on an AFTC Form 5001, or equivalent, in accordance with **Chapter 6**.

8.4.1. **(Added-96TWSUP)** Project Managers (or equivalent) will ensure their programs are reviewed IAW AFTCI 91-202 timelines and capture lessons learned as necessary during this review. Time reviews should include appropriate stakeholders and those inputs consolidated and archived with test documentation. This review will be coordinated with 96 TW/SEU. Documentation of this review will be archived with all test documentation.

8.5. Closure Amendments/Lessons Learned. A closure amendment or close-out notification email provided by the test team may be used to notify the test safety office that the existing safety plan is no longer in use. Closure amendments can be used to document lessons learned over the course of the test project (see section 2.1.7.8) or other formats as specified in local supplements to this instruction. A well-written closure amendment could close the loop on a test package by re-assessing the GMPs and THAs. In addition, it could help future researchers benefit from the lessons learned during testing and obtain pertinent information that the test team would have liked to know at the beginning of the test project. Lessons learned can also be captured when the three year time limit has been reached.

8.5.1. **(Added-96TWSUP)** Lessons learned may be captured in the format shown in **Attachment 5;** contact 96 TW/SEU for a sample worksheet. The author of the closeout letter is responsible for requesting any lessons learned from personnel supporting the activity and consolidating the responses as an attachment to the closeout letter. The closeout letter will be archived with all test documentation IAW 96 TWI 99-100 or similar instruction.

CHRISTOPHER P. AZZANO,
Brigadier General, USAF
Commander

(96TWSUP)

SCOTT A. CAIN,
Brigadier General, USAF
Commander

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFI 11-215, *USAF Flight Manuals Program (FMP),* 3 January 2011

AFI 11-215_AFMCSUP1, *USAF Flight Manuals Program (FMP)*, 25 May 2011

AFI 33-360, *Publications and Forms Management,* 25 September 2013

AFI 91-202, *The US Air Force Mishap Prevention Program*, 24 June 2015

AFI 91-202_AFMCSUP, *The US Air Force Mishap Prevention Program*, 17 May 2017

AFI 91-204_AFMCSUP, *Safety Investigations and Reports,* 23 March 2016

AFI 99-103, *Capabilities-Based Test and Evaluation*, 6 April 2017

(Added-96TWSUP) Air Force Instruction (AFI) 33-322, *Records Management and Information Governance Program*

AFMAN 33-363, *Management of Records,* 01 March 2008

AFPAM 90-803, *Risk Management (RM) Guidelines and Tools*, 11 February 2013

AFTCI 62-602, *Developmental Engineering, Airworthiness,* 18 February 2015

MIL-STD-882E, *Standard Practice for System Safety*, 11 May 2012

Prescribed Forms

AFTC Form 5000, *Test Hazard Analysis*

AFTC Form 5001, *Test Project Safety Review*

Adopted Forms

AF Form 847, *Recommendation for Change of Publication*

AF Form 813, *Request for Environmental Impact Analysis*

AFTC Form 6239, *T-2 Modification Airworthiness Compliance*

Abbreviations and Acronyms

AEDC—Arnold Engineering Development Complex

AFI—Air Force Instruction

AFMAN—Air Force Manual

AFMC—Air Force Materiel Command

AFRIMS—Air Force Records Information Management System

(Added-96TWSUP) AFSEO—Air Force SEEK EAGLE Office

AFTC—Air Force Test Center

BHA—Baseline Hazard Analysis

BSR—Baseline Safety Report

CCB—Configuration Control Board

(Added-96TWSUP) CFP—Compatibility Flight Profile

(Added-96TWSUP) CONOPS—Concept of Operations

DRB—Design Review Board

ESR—Electronic Safety Review

ESR—Electronic Safety Review

ETO—Execution Test Organization

(Added-96TWSUP) ETTC—Eglin Test and Training Complex

GMP—General Minimizing Procedures

(Added-96TWSUP) GWEF—Guided Weapons Evaluation Facility

IAW—In Accordance With

ISR—Independent Safety Reviewers

(Added-96TWSUP) ISR—Independent Safety Reviewer

(Added-96TWSUP) JON—Job Order Number

(Added-96TWSUP) J-PRIMES—Joint Preflight Integration of Munitions and Electronic Systems

LDTO—Lead Developmental Test Organization

MOT—Method of Test

NRR—Negligible Risk Review

NRR—Negligible Risk Review

(Added-96TWSUP) OPR—Office of Primary Responsibility

(Added-96TWSUP) PTO—Participating Test Organization

RDS—Records Disposition Schedule

RM—Risk Management

(Added-96TWSUP) SA—Safety Appendix

SE—Safety Office

SOC—Statement of Capability

(Added-96TWSUP) SPO—System Program Office

SRB—Safety Review Board

SRBS—Safety Review Board Summary

SUT—System Under Test

T&E—Test and Evaluation

T.O.—Technical Order

(Added-96TWSUP) T/RSE—Test/Range Safety Engineer

TAB—Test Approval Brief

(Added-96TWSUP) TD—Test Directive

(Added-96TWSUP) TE—Test Engineer

TEA—Test Execution Authority

THA—Test Hazard Analysis

TIS—Test Information Sheet

(Added-96TWSUP) TM—Test Manager

TRB—Technical Review Board

TRB—Technical Review Board

TSO—Test Safety Officer

TW—Test Wing

(Added-96TWSUP) VTC—Video Teleconference

Terms

Acceptable Risk—That part of identified risk which is allowed by the managing activity to persist without further engineering or management action.

Baseline Hazard Analyses (BHA) —An analysis used to document known hazards concerned with the normal day-to-day operation and maintenance of a test system, subsystem or ground test facility.

Baseline Safety Report (BSR)—A compilation of BHAs that constitute the hazards associated with the specific operation of a test system, subsystem or ground test facility and includes a BHA for all systems to be operated or maintained. The BSR allows the individual hazard analyses that make up the baseline to be evaluated in a comprehensive package and thus shows the interaction of the systems and interfaces.

Control/Safety Measure—An action taken to eliminate or reduce a potential test hazard to a more acceptable risk level.

General Minimizing Procedure (GMP) —Statements that direct a specific action or procedure that mitigates general test execution risk; these generally include the words “will” or “shall”. GMPs are used to address system under test restrictions, test build-up, critical parameter monitoring, go-no-go criteria, weather or environmental criteria, and flight test chase requirements among other items of test safety concern.

Hazard—Any real or potential condition that can cause injury, illness, or death to personnel; damage to or loss of a system, equipment or property; or damage to the environment. It is the threat of harm and is a precursor state to a mishap.

Identified Risk—That risk which has been determined through various analysis techniques.

Independent Review—A review by an individual or group that does not have a vested interest in the successful accomplishment of the test objectives and was not directly responsible for the development of the test package.

Initial risk—The first assessment of the potential risk of an identified hazard. Initial risk establishes a fixed baseline for the hazard.

Mishap—An unplanned event or series of events resulting in death, injury, occupational illness, or damage to or loss of equipment or property, or damage to the environment, and meets Class A, B, C or D reporting criteria IAW AFI 91-204.

Mishap Accountability—The identification of an "owning unit or units" (see AFI 91-204 for definition) of the mishap assets and/or personnel and the unit assuming investigative responsibility (Convening Authority) if other than the owning unit or if multiple owning units are involved. Mishap accountability must be established IAW AFI 91-204 prior to conducting tests per AFI 99-103.

Mitigation Measure—Action required to eliminate the hazard or when a hazard cannot be eliminated, reduce the associated risk by lessening the severity of the resulting mishap or lowering the likelihood that a mishap will occur. (MIL-STD-882). These are also referred to as a countermeasure or a control/safety measure and can be captured as a GMP.

Probability—An expression of the likelihood of occurrence of a mishap.

Residual Risk—The remaining mishap risk that exists after all mitigation measures have been implemented or exhausted, in accordance with the system safety design order of precedence.

Risk—A combination of the severity of the mishap and the probability that the mishap will occur.

Risk Level—An expression of the danger posed by a hazard in terms of the severity of outcome and the probability of occurrence. Risk level is assigned to a hazard or to a combination of hazards. As such, risk levels are assigned to both a test event and the test as a whole.

Risk Management (RM)—The systematic process of identifying threats/hazards/problems, assessing risk, analyzing risk control options and measures, making control decisions, implementing control decisions, accepting residual risks, and supervising/reviewing the activity for effectiveness.

Safety Plan—Safety documentation that details the specific safety criteria and parameters to allow safe conduct of a test. The safety plan can identify targets, munitions, aircraft, and other equipment to be used; defines danger areas; identifies the potential hazards associated with the test; and establishes the specific safety requirements necessary to conduct the test, such as special handling, flight termination systems, surveillance requirements, communication requirements, etc.

Safety Review Board—An independent panel of subject knowledgeable individuals that review the test and associated safety plan to ensure test hazards are identified; then eliminated, minimized or controlled to an acceptable level; and to establish the overall risk level. As a minimum, the safety reviewer panel will be composed of a technical and operations representative who will review the test package. Technical representatives are chosen based on their experience and expertise in the engineering discipline(s) associated with the test activity to be reviewed. Operations representatives are chosen based on their test and operations experience in similar test activities. An SRB chair will be appointed as one of the safety reviewers. Other independent reviewers can include range safety, maintenance, logistics, etc. as appropriate for the test.

Severity—The magnitude of potential consequences of a mishap to include: death, injury, occupational illness, damage to or loss of equipment or property, damage to the environment, or monetary loss. Damage to the environment will be assessed through the appropriate channels and documented on an AF Form 813.

System Safety—The application of engineering and management principles, criteria, and techniques to achieve acceptable risk within the constraints of operational effectiveness and suitability, time, and cost throughout all phases of the system life-cycle. (MIL-STD-882)

Test Execution Authority (TEA)—The individual responsible for accepting the SRB and Technical Review Board results and approving the test to proceed with any residual risk.

Test Hazard Analysis (THA)—A document that identifies test unique hazards, causes, and effects and establishes controls which are used to determine risk level. For AFTC test projects, test hazard analysis will be documented on an AFTC Form 5000 or equivalent.

Test Organization/Unit—The organization or unit providing the test facilities, equipment or personnel to conduct a test. The system under test may or may not be a resource of the test organization/unit. Also known as the executing test organization (ETO).

Test Organization/Unit Commander—The highest ranking individual at the test organization or unit (commander or director). This individual has responsibility for the personnel, equipment and/or facilities for accomplishing the test, and is the individual responsible for reporting mishaps involving the system under test or the facilities.

Test Package—As a minimum, the test package includes the test plan, safety plan and any other appendices or documentation that supports the test planning.

Test Plan—The test plan describes the system under test, defines the test objectives and outlines the test methodology in sufficient detail to demonstrate technical adequacy and execute a technically effective test project.

Test Safety—The application of engineering and management principles, criteria, and techniques to optimize all aspects of safety within the constraints of operational effectiveness, time and cost throughout the defined test cycle.

Test Safety Office—The safety office responsible for oversight and support of the Lead Developmental Test Organization. This responsibility may reside in the test organization's safety office or the Center/Installation safety office.

Test Unique Hazards—Hazards that are a result of the specific test being accomplished and not present in the normal operational hazards associated with the system or environment. These hazards include those inherent to the article being tested as well as those hazards associated with the testing of any systems.

Unacceptable Risk—That risk which cannot be tolerated by the managing activity. It is a subset of identified risk. Unacceptable risk is either eliminated or controlled.

Waiver—Approval from the appropriate authority to deviate from both the intent and the letter of the requirement.

Attachment 2 (Added-96TWSUP)

SAMPLE SAFETY APPENDIX FORMAT

Figure A2.1. (Added-96TWSUP) Sample Safety Appendix Format.

SAFETY APPENDIX
AMENDMENT NO. X (if applicable)
TEST DIRECTIVE NO. XXXXXXXX
TEST DIRECTIVE AMENDMENT NO. X (if applicable)
PROJECT/TEST TITLE

1. Purpose. *(Briefly discuss the test. Include whether items are live or inert, how many sorties/missions, etc; keep it to one or two sentences.)* The following safety criteria have been established for conduct of the test. The overall risk level for this test is **High**, **Medium**, or **Low**.

2. Test Areas, Test Aircraft, Test Items (and/or munitions) and Target(s).
a. Test Areas:
b. Test Aircraft:
c. Test Items:
d. Target(s):

3. Danger Areas.
a. (Describe danger area)
b. (Another danger area)

4. Potential Hazards. A Safety Review Board (SRB) was convened on DD MMM YYYY to identify potential hazards and discuss high interest areas. The results of that meeting can be found in the SRB Summary (SRBS) as an attachment to this Safety Appendix. Test-specific hazards identified with their post-mitigation residual risk are: (*Hazards are ordered from **High** to **Low**)*

a. Hazard one *(identify this hazard as either* high, medium, *or* low*)*
b. Hazard two *(identify this hazard as either* high, medium, *or* low*)*

5. Mishap Accountability. Mishap accountability will be IAW AFI 91-204. (*If the aircraft is contractor-owned and contractor-operated, then mishap accountability will be with the contractor.) Include any discussion when deviation from AFI 91-204 is requested.*

6. Safety Requirements. The test team proposed an initial safety plan. The safety plan was reviewed, discussed, and amended. The final safety plan is detailed below.
a. General Mitigating Requirements
(1) (Requirements)
(2) (Requirements)

b. Hazard One
(1) (Requirements)
(2) (Requirements)

7. Range Safety Instrumentation Requirements.

Atch: SRBS

Prepared by:	Approved by:
//Digitally Signed//	//Digitally Signed//
SRB Chairperson Name Test and Range Safety	SA Approval Name (Deputy) Director of Safety

Attachment 3 (Added-96TWSUP)

INSTRUCTIONS FOR COMPLETING AFTC FORM 5000, TEST HAZARD ANALYSIS FORM

Figure A3.1. (Added-96TWSUP) Instructions for Completing AFTC Form 5000, Test Hazard Analysis Form.

GENERAL. The Test Hazard Analysis (THA) form is to facilitate discussion of the identified hazards by providing a concise way to become familiar with your test. The lower section of the form is designed to be filled out prior to the SRB--usually by the test engineer. Risk levels will be assigned during the SRB. Each form should address a single hazard, although it may contain any number of causes, effects, minimizing procedures and emergency procedures. Please do not address unquestionably "Low Risk" hazards that require no minimizing procedures or hazards intrinsic to aircraft or basic equipment operation.

"HAZARD" is defined as something TEST UNIQUE. It's a CONDITION that has the potential to cause a mishap or an accident resulting in an unfavorable end to the test. For example, a hazard might be the early, unintended detonation of test munitions.

"CAUSE" is defined as the CIRCUMSTANCES or SITUATION that lead to the hazard's occurrence. For example, early detonation might be caused by the proximity of heavy equipment or a chase plane.

"EFFECT" is defined as the MISHAP or ACCIDENT that we are trying to avoid. The EFFECT will determine the MISHAP SEVERITY CATEGORY as defined below. In our running example, the effect might be the loss of equipment, aircraft or lives.

"MINIMIZING PROCEDURES" are the things you do to prevent the hazard from occurring or reduce its probability and/or severity. For instance, you may specify a minimum clearance distance from the munitions.

"EMERGENCY PROCEDURES" are test specific and fairly self-explanatory. Not all hazards will have appropriate emergency procedures.

"COMMENTS" are made by anyone who wishes to record thoughts or considerations: the SRB, Safety, TAB, etc. These may also document risk reduction activities that are performed prior to the test for which the hazard has been identified.

RISK LEVEL ASSESSMENT. The risk level is assessed by breaking the hazard into two parts: How much damage might result, and how probable it may be. The amount of damage is categorized according to EFFECT and is fairly straightforward. The probability of occurrence is highly subjective and may be difficult to agree upon. The guidelines below may be of some help.

MISHAP SEVERITY CATEGORY.

CATASTROPHIC: Loss of life, aircraft, facility or expensive system. Loss > $2,500,000 (e.g., Aircraft Class A).

CRITICAL: Severe injury, lengthy hospital stay, permanent injury. Severe aircraft, equipment or property damage. Loss > $600,000 but < $2,500,000 (e.g., Aircraft Class B).

MARGINAL: Minor injury, requiring medical lost work days, but no permanent injury. Minor damage. Loss > $60,000 but < $600,000.

NEGLIGIBLE: Superficial injury, little or no first aide required. Incidental, less than minor damage. Loss < $60,000.

SUBJECTIVE PROBABILITY OF MISHAP.

FREQUENT: Very likely to occur (e.g., test exceeds design limits or mishap occurred during similar testing).

PROBABLE: Likely to occur (e.g., test at design limits or mishap almost occurred during similar testing).

OCCASIONAL: Some likelihood to occur, but not expected.

REMOTE: Unlikely to occur (e.g., test activity within design limits and done before w/no problems encountered).

IMPROBABLE: There just is not a problem. Nothing has ever gone wrong. Several failures required to have any serious consequences.

Attachment 4 (Added-96TWSUP)

SAMPLE SAFETY REVIEW BOARD SUMMARY

Figure A4.1. (Added-96TWSUP) Sample Safety Review Board Summary.

SAFETY REVIEW BOARD SUMMARY

TEST DIRECTIVE NO. XXXXXXXX

TEST DIRECTIVE NO. X *(if applicable)*

PROJECT/TEST TITLE

1. Safety Review Board Date: DD MMM YYYY

2. High Risk Hazards: Hazard *(or None)*

3. Medium Risk Hazards: Hazard *(or None)*

4. Low Risk Hazards:
a. Hazard
b. Hazard

5. SRB Members:

Name	Organization	Telephone
SRB Chairperson Name*	96 TW/SEU	xxx-xxx-xxxx
Rank First Last		

*SRB Chairperson
**via telecon (if applicable)

6. Background:

7. Design Safety Certifications. All hazardous items to be utilized during this test have been evaluated with respect to design safety *(Identify who evaluated any hazardous items, if applicable)*. These items are considered to present no unusual or unacceptable hazards during the testing described in the Test Directive.

8. Safety Footprint Methodology. *(Explain how you came up with the size/location of the footprint and hazard distances. Were data provided by contractors, the SPO, or someone else? This section correlates with Para 3 of the SA)*

9. Discussion Items.

a. Hazard One (Severity – Probability). *(Relevant discussion about the hazard. Should include causes, effects, minimizing procedures and the risk assessment based on severity and probability)*

b. Hazard Two (Severity – Probability). *(Relevant discussion about the hazard. Should include causes, effects, minimizing procedures and the risk assessment based on severity and probability)*

c. Other Discussion Items:
(1) (e.g., Required flight clearances…)

(2) (e.g., Weather restrictions…)

(3) (Other notable discussion topics)

Atch: THA Forms

Prepared by:	Approved by:
//Digitally Signed//	//Digitally Signed//
SRB Chairperson Name	SRBS Approval Name
Test and Range Safety	Chief of Test and Range Safety

Attachment 5 (Added-96TWSUP)

SAMPLE LESSONS LEARNED WORKSHEET

Figure A5.1. (Added-96TWSUP) Sample Lessons Learned Worksheet.

PROJECT TEST TITLE

JON NUMBER

SYSTEM

TEST POINT(S) OR CONDITION(S)

HAZARD(S) [IF APPLICABLE]

LESSON(S) LEARNED

EXISTING (OLD) PROCEDURE(S)

NEW OR AMENDED PROCEDURE(S)

COMMENTS